



BOMB DEFUSAL MANUAL

Version 1

Verification Code: 241

Welcome to the dangerous and challenging world of bomb defusing.

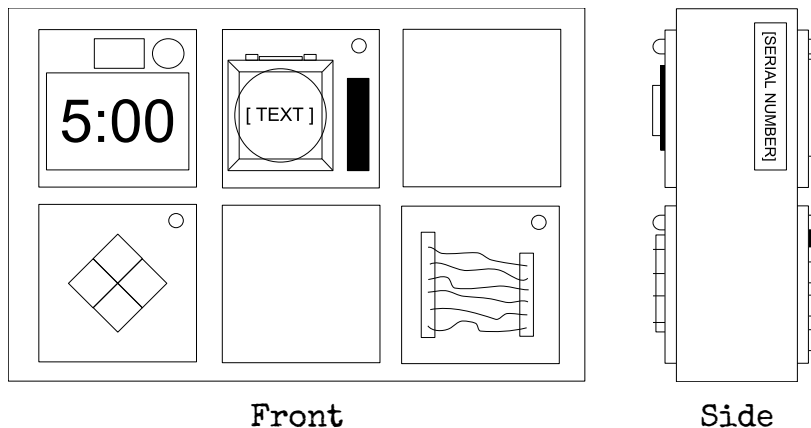
Study this manual carefully; you are the expert. In these pages you will find everything you need to know to defuse even the most insidious of bombs.

And remember — One small oversight and it could all be over!

Defusing Bombs

A bomb will explode when its countdown timer reaches 0:00 or when too many strikes have been recorded. The only way to defuse a bomb is to disarm all of its modules before its countdown timer expires.

Example Bomb



Modules

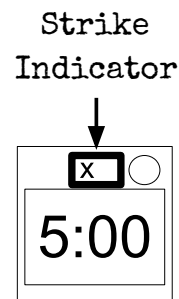
Each bomb will include up to 11 modules that must be disarmed. Each module is discrete and can be disarmed in any order.

Instructions for disarming modules can be found in Section 1. "Needy" modules present a special case and are described in Section 2.

Strikes

When the Defuser makes a mistake the bomb will record a strike which will be displayed on the indicator above the countdown timer. Bombs with a strike indicator will explode upon the third strike. The timer will begin to count down faster after a strike has been recorded.

If no strike indicator is present above the countdown timer, the bomb will explode upon the first strike, leaving no room for error.



Gathering Information

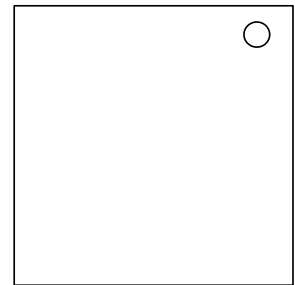
Some disarming instructions will require specific information about the bomb, such as the serial number. This type of information can typically be found on the top, bottom, or sides of the bomb casing. See Appendix A, B, and C for identification instructions that will be useful in disarming certain modules.

Section 1: Modules

Modules can be identified by an LED in the top right corner.

When this LED is lit green the module has been disarmed.

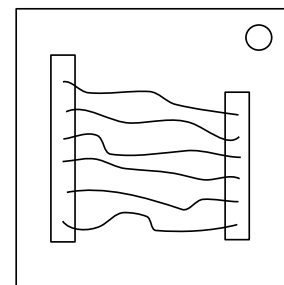
All modules must be disarmed to defuse the bomb.



On the Subject of Wires

*Wires are the lifeblood of electronics! Wait, no, electricity is the lifeblood.
Wires are more like the arteries. The veins? No matter...*

- A wire module can have 3-6 wires on it.
- Only the one correct wire needs to be cut to disarm the module.
- Wire ordering begins with the first on the top.



3 wires:

If there are no red wires, cut the second wire.

Otherwise, if the last wire is white, cut the last wire.

Otherwise, if there is more than one blue wire, cut the last blue wire.

Otherwise, cut the last wire.

4 wires:

If there is more than one red wire and the last digit of the serial number is odd, cut the last red wire.

Otherwise, if the last wire is yellow and there are no red wires, cut the first wire.

Otherwise, if there is exactly one blue wire, cut the first wire.

Otherwise, if there is more than one yellow wire, cut the last wire.

Otherwise, cut the second wire.

5 wires:

If the last wire is black and the last digit of the serial number is odd, cut the fourth wire.

Otherwise, if there is exactly one red wire and there is more than one yellow wire, cut the first wire.

Otherwise, if there are no black wires, cut the second wire.

Otherwise, cut the first wire.

6 wires:

If there are no yellow wires and the last digit of the serial number is odd, cut the third wire.

Otherwise, if there is exactly one yellow wire and there is more than one white wire, cut the fourth wire.

Otherwise, if there are no red wires, cut the last wire.

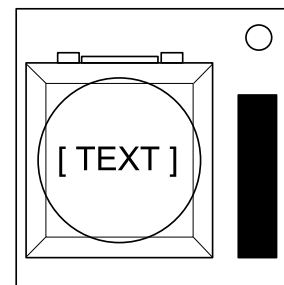
Otherwise, cut the fourth wire.

On the Subject of The Button

You might think that a button telling you to press it is pretty straightforward. That's the kind of thinking that gets people exploded.

See Appendix A for indicator identification reference.

See Appendix B for battery identification reference.



Follow these rules in the order they are listed. Perform the first action that applies:

1. If the button is blue and the button says "Abort", hold the button and refer to "Releasing a Held Button".
2. If there is more than 1 battery on the bomb and the button says "Detonate", press and immediately release the button.
3. If the button is white and there is a lit indicator with label CAR, hold the button and refer to "Releasing a Held Button".
4. If there are more than 2 batteries on the bomb and there is a lit indicator with label FRK, press and immediately release the button.
5. If the button is yellow, hold the button and refer to "Releasing a Held Button".
6. If the button is red and the button says "Hold", press and immediately release the button.
7. If none of the above apply, hold the button and refer to "Releasing a Held Button".

Releasing a Held Button

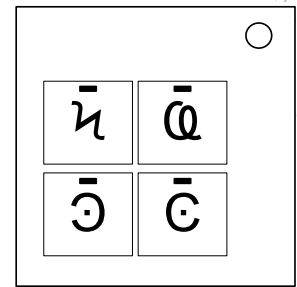
If you start holding the button down, a colored strip will light up on the right side of the module. Based on its color you must release the button at a specific point in time:

- Blue strip: release when the countdown timer has a 4 in any position.
- White strip: release when the countdown timer has a 1 in any position.
- Yellow strip: release when the countdown timer has a 5 in any position.
- Any other color strip: release when the countdown timer has a 1 in any position.

On the Subject of Keypads

I'm not sure what these symbols are, but I suspect they have something to do with occult.

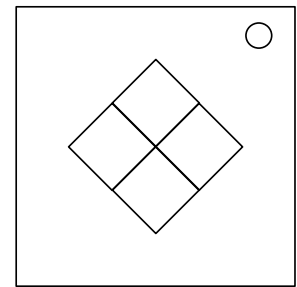
- Only one column below has all four of the symbols from the keypad.
- Press the four buttons in the order their symbols appear from top to bottom within that column.



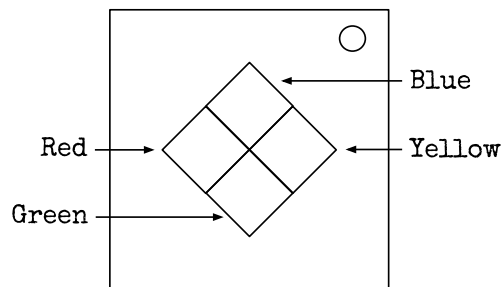
Q	Ë	©	б	Ψ	б
A	Q	ٲ	¶	ٲ	Ë
λ	⊖	Q	Ђ	Ђ	✕
h	Q	Ж	ИЖ	С	æ
ИЖ	☆	¿	Ж	¶	Ψ
х	х	λ	¿	з	Й
⊖	¿	☆	ٲ	★	Ω

On the Subject of Simon Says

This is like one of those toys you played with as a kid where you have to match the pattern that appears, except this one is a knockoff that was probably purchased at a dollar store.



1. One of the four colored buttons will flash.
2. Using the correct table below, press the button with the corresponding color.
3. The original button will flash, followed by another. Repeat this sequence in order using the color mapping.
4. The sequence will lengthen by one each time you correctly enter a sequence until the module is disarmed.



If the serial number contains a vowel:

		Red Flash	Blue Flash	Green Flash	Yellow Flash
Button to press:	No Strikes	Blue	Red	Yellow	Green
	1 Strike	Yellow	Green	Blue	Red
	2 Strikes	Green	Red	Yellow	Blue

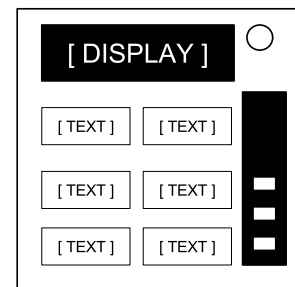
If the serial number does not contain a vowel:

		Red Flash	Blue Flash	Green Flash	Yellow Flash
Button to press:	No Strikes	Blue	Yellow	Green	Red
	1 Strike	Red	Blue	Yellow	Green
	2 Strikes	Yellow	Green	Blue	Red

On the Subject of Who's on First

This contraption is like something out of a sketch comedy routine, which might be funny if it wasn't connected to a bomb. I'll keep this brief, as words only complicate matters.

1. Read the display and use step 1 to determine which button label to read.
2. Using this button label, use step 2 determine which button to push.
3. Repeat until the module has been disarmed.



Step 1:

Based on the display, read the label of a particular button and proceed to step 2:

YES	FIRST	DISPLAY	OKAY	SAYS	NOTHING
BLANK	NO	LED	LEAD	READ	
RED	REED	LEED	HOLD ON	YOU	YOU ARE
YOUR	YOU'RE	UR	THERE	THEY'RE	THEIR
THEY ARE	SEE	C	CEE		

Step 2:

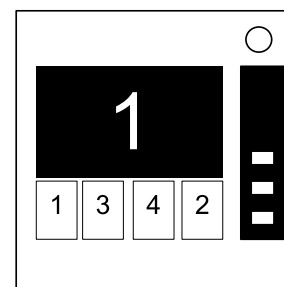
Using the label from step 1, push the first button that appears in its corresponding list:

"READY":	YES, OKAY, WHAT, MIDDLE, LEFT, PRESS, RIGHT, BLANK, READY, NO, FIRST, UHHH, NOTHING, WAIT
"FIRST":	LEFT, OKAY, YES, MIDDLE, NO, RIGHT, NOTHING, UHHH, WAIT, READY, BLANK, WHAT, PRESS, FIRST
"NO":	BLANK, UHHH, WAIT, FIRST, WHAT, READY, RIGHT, YES, NOTHING, LEFT, PRESS, OKAY, NO, MIDDLE
"BLANK":	WAIT, RIGHT, OKAY, MIDDLE, BLANK, PRESS, READY, NOTHING, NO, WHAT, LEFT, UHHH, YES, FIRST
"NOTHING":	UHHH, RIGHT, OKAY, MIDDLE, YES, BLANK, NO, PRESS, LEFT, WHAT, WAIT, FIRST, NOTHING, READY
"YES":	OKAY, RIGHT, UHHH, MIDDLE, FIRST, WHAT, PRESS, READY, NOTHING, YES, LEFT, BLANK, NO, WAIT
"WHAT":	UHHH, WHAT, LEFT, NOTHING, READY, BLANK, MIDDLE, NO, OKAY, FIRST, WAIT, YES, PRESS, RIGHT
"UHHH":	READY, NOTHING, LEFT, WHAT, OKAY, YES, RIGHT, NO, PRESS, BLANK, UHHH, MIDDLE, WAIT, FIRST
"LEFT":	RIGHT, LEFT, FIRST, NO, MIDDLE, YES, BLANK, WHAT, UHHH, WAIT, PRESS, READY, OKAY, NOTHING
"RIGHT":	YES, NOTHING, READY, PRESS, NO, WAIT, WHAT, RIGHT, MIDDLE, LEFT, UHHH, BLANK, OKAY, FIRST
"MIDDLE":	BLANK, READY, OKAY, WHAT, NOTHING, PRESS, NO, WAIT, LEFT, MIDDLE, RIGHT, FIRST, UHHH, YES
"OKAY":	MIDDLE, NO, FIRST, YES, UHHH, NOTHING, WAIT, OKAY, LEFT, READY, BLANK, PRESS, WHAT, RIGHT
"WAIT":	UHHH, NO, BLANK, OKAY, YES, LEFT, FIRST, PRESS, WHAT, WAIT, NOTHING, READY, RIGHT, MIDDLE
"PRESS":	RIGHT, MIDDLE, YES, READY, PRESS, OKAY, NOTHING, UHHH, BLANK, LEFT, FIRST, WHAT, NO, WAIT
"YOU":	SURE, YOU ARE, YOUR, YOU'RE, NEXT, UH HUH, UR, HOLD, WHAT?, YOU, UH UH, LIKE, DONE, U
"YOU ARE":	YOUR, NEXT, LIKE, UH HUH, WHAT?, DONE, UH UH, HOLD, YOU, U, YOU'RE, SURE, UR, YOU ARE
"YOUR":	UH UH, YOU ARE, UH HUH, YOUR, NEXT, UR, SURE, U, YOU'RE, YOU, WHAT?, HOLD, LIKE, DONE
"YOU'RE":	YOU, YOU'RE, UR, NEXT, UH UH, YOU ARE, U, YOUR, WHAT?, UH HUH, SURE, DONE, LIKE, HOLD
"UR":	DONE, U, UR, UH HUH, WHAT?, SURE, YOUR, HOLD, YOU'RE, LIKE, NEXT, UH UH, YOU ARE, YOU
"U":	UH HUH, SURE, NEXT, WHAT?, YOU'RE, UR, UH UH, DONE, U, YOU, LIKE, HOLD, YOU ARE, YOUR
"UH HUH":	UH HUH, YOUR, YOU ARE, YOU, DONE, HOLD, UH UH, NEXT, SURE, LIKE, YOU'RE, UR, U, WHAT?
"UH UH":	UR, U, YOU ARE, YOU'RE, NEXT, UH UH, DONE, YOU, UH HUH, LIKE, YOUR, SURE, HOLD, WHAT?
"WHAT?":	YOU, HOLD, YOU'RE, YOUR, U, DONE, UH UH, LIKE, YOU ARE, UH HUH, UR, NEXT, WHAT?, SURE
"DONE":	SURE, UH HUH, NEXT, WHAT?, YOUR, UR, YOU'RE, HOLD, LIKE, YOU, U, YOU ARE, UH UH, DONE
"NEXT":	WHAT?, UH HUH, UH UH, YOUR, HOLD, SURE, NEXT, LIKE, DONE, YOU ARE, UR, YOU'RE, U, YOU
"HOLD":	YOU ARE, U, DONE, UH UH, YOU, UR, SURE, WHAT?, YOU'RE, NEXT, HOLD, UH HUH, YOUR, LIKE
"SURE":	YOU ARE, DONE, LIKE, YOU'RE, YOU, HOLD, UH HUH, UR, SURE, U, WHAT?, NEXT, YOUR, UH UH
"LIKE":	YOU'RE, NEXT, U, UR, HOLD, DONE, UH UH, WHAT?, UH HUH, YOU, LIKE, SURE, YOU ARE, YOUR

On the Subject of Memory

Memory is a fragile thing but so is everything else when a bomb goes off, so pay attention!

- Press the correct button to progress the module to the next stage. Complete all stages to disarm the module.
- Pressing an incorrect button will reset the module back to stage 1.
- Button positions are ordered from left to right.



Stage 1:

If the display is 1, press the button in the second position.

If the display is 2, press the button in the second position.

If the display is 3, press the button in the third position.

If the display is 4, press the button in the fourth position.

Stage 2:

If the display is 1, press the button labeled "4".

If the display is 2, press the button in the same position as you pressed in stage 1.

If the display is 3, press the button in the first position.

If the display is 4, press the button in the same position as you pressed in stage 1.

Stage 3:

If the display is 1, press the button with the same label you pressed in stage 2.

If the display is 2, press the button with the same label you pressed in stage 1.

If the display is 3, press the button in the third position.

If the display is 4, press the button labeled "4".

Stage 4:

If the display is 1, press the button in the same position as you pressed in stage 1.

If the display is 2, press the button in the first position.

If the display is 3, press the button in the same position as you pressed in stage 2.

If the display is 4, press the button in the same position as you pressed in stage 2.

Stage 5:

If the display is 1, press the button with the same label you pressed in stage 1.

If the display is 2, press the button with the same label you pressed in stage 2.

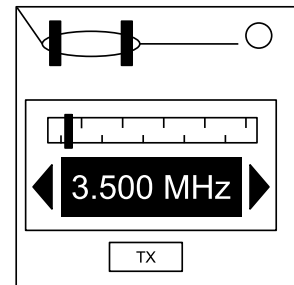
If the display is 3, press the button with the same label you pressed in stage 4.

If the display is 4, press the button with the same label you pressed in stage 3.

On the Subject of Morse Code

An antiquated form of naval communication? What next? At least it's genuine Morse Code, so pay attention and you might just learn something.

- Interpret the signal from the flashing light using the Morse Code chart to spell one of the words in the table.
- The signal will loop, with a long gap between repetitions.
- Once the word is identified, set the corresponding frequency and press the transmit (TX) button.



How to Interpret

1. A short flash represents a dot.
2. A long flash represents a dash.
3. There is a long gap between letters.
4. There is a very long gap before the word repeats.

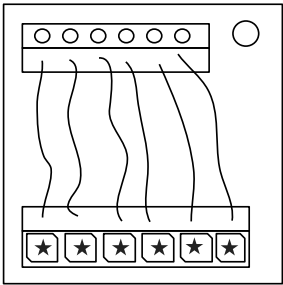
A ● ■■■	U ● ● ■■■
B ■■■ ● ● ●	V ● ● ● ■■■
C ■■■ ● ■■■ ●	W ● ■■■ ■■■
D ■■■ ● ●	X ■■■ ● ● ■■■
E ●	Y ■■■ ● ■■■ ■■■
F ● ● ■■■ ●	Z ■■■ ■■■ ● ●
G ■■■ ■■■ ●	
H ● ● ● ●	
I ● ●	
J ● ■■■ ■■■ ■■■	
K ■■■ ● ■■■	
L ● ■■■ ● ●	
M ■■■ ■■■	
N ■■■ ●	
O ■■■ ■■■ ■■■	
P ● ■■■ ■■■ ●	
Q ■■■ ■■■ ● ■■■	
R ● ■■■ ●	
S ● ● ●	
T ■■■	

1 ● ■■■ ■■■ ■■■ ■■■
2 ● ● ■■■ ■■■ ■■■
3 ● ● ● ■■■ ■■■
4 ● ● ● ● ■■■
5 ● ● ● ● ●
6 ■■■ ● ● ● ●
7 ■■■ ■■■ ● ● ●
8 ■■■ ■■■ ■■■ ● ●
9 ■■■ ■■■ ■■■ ■■■ ●
0 ■■■ ■■■ ■■■ ■■■ ■■■

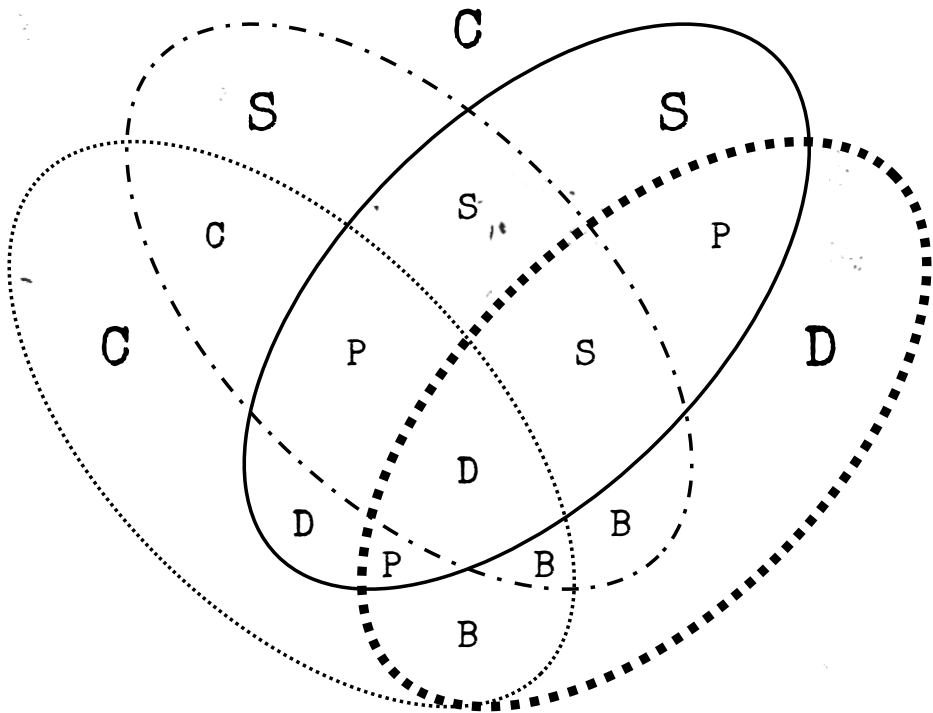
If the word is:	Respond at frequency:
shell	3.505 MHz
halls	3.515 MHz
slick	3.522 MHz
trick	3.532 MHz
boxes	3.535 MHz
leaks	3.542 MHz
strobe	3.545 MHz
bistro	3.552 MHz
flick	3.555 MHz
bombs	3.565 MHz
break	3.572 MHz
brick	3.575 MHz
steak	3.582 MHz
sting	3.592 MHz
vector	3.595 MHz
beats	3.600 MHz

On the Subject of Complicated Wires

These wires aren't like the others. Some have stripes! That makes them completely different. The good news is that we've found a concise set of instructions on what to do about it! Maybe too concise...



- Look at each wire: there is an LED above the wire and a space for a "★" symbol below the wire.
- For **each** wire/LED/symbol combination, use the Venn diagram below to decide whether or not to cut the wire.
- Each wire may be striped with multiple colors.



	Wire has red coloring
	Wire has blue coloring
	Has ★ symbol
	LED is on

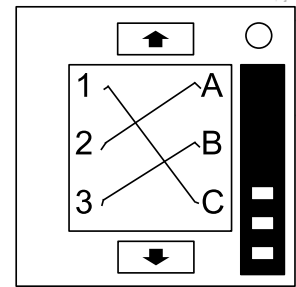
Letter	Instruction
C	Cut the wire
D	Do not cut the wire
S	Cut the wire if the last digit of the serial number is even
P	Cut the wire if the bomb has a parallel port
B	Cut the wire if the bomb has two or more batteries

See Appendix B for battery identification reference.
See Appendix C for port identification reference.

On the Subject of Wire Sequences

It's hard to say how this mechanism works. The engineering is pretty impressive, but there must have been an easier way to manage nine wires.

- Within this module there are several panels with wires on them, but only one panel is visible at a time. Switch to the next panel by using the down button and the previous panel by using the up button.
- Do not switch to the next panel until you are sure that you have cut all necessary wires on the current panel.
- Cut the wires as directed by the following table. Wire occurrences are cumulative over all panels within the module.

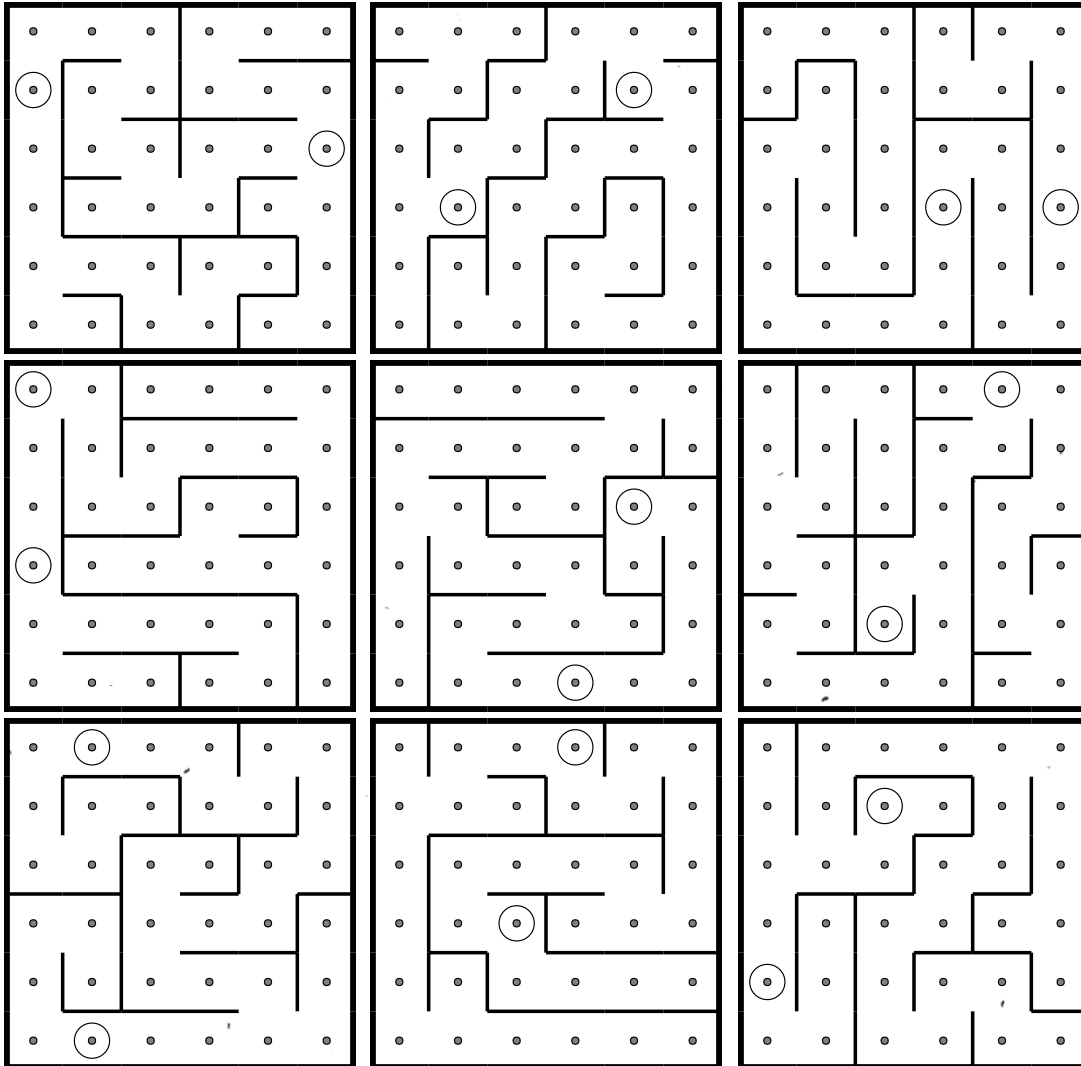
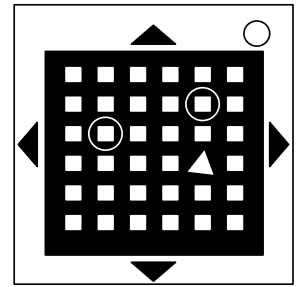


Red Wire Occurrences		Blue Wire Occurrences		Black Wire Occurrences	
Wire Occurrence	Cut if connected to:	Wire Occurrence	Cut if connected to:	Wire Occurrence	Cut if connected to:
First red occurrence	C	First blue occurrence	B	First black occurrence	A, B or C
Second red occurrence	B	Second blue occurrence	A or C	Second black occurrence	A or C
Third red occurrence	A	Third blue occurrence	B	Third black occurrence	B
Fourth red occurrence	A or C	Fourth blue occurrence	A	Fourth black occurrence	A or C
Fifth red occurrence	B	Fifth blue occurrence	B	Fifth black occurrence	B
Sixth red occurrence	A or C	Sixth blue occurrence	B or C	Sixth black occurrence	B or C
Seventh red occurrence	A, B or C	Seventh blue occurrence	C	Seventh black occurrence	A or B
Eighth red occurrence	A or B	Eighth blue occurrence	A or C	Eighth black occurrence	C
Ninth red occurrence	B	Ninth blue occurrence	A	Ninth black occurrence	C

On the Subject of Mazes

This seems to be some kind of maze, probably stolen off of a restaurant placemat.

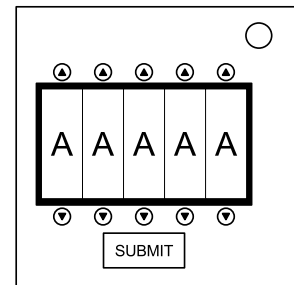
- Find the maze with matching circular markings.
- The defuser must navigate the white light to the red triangle using the arrow buttons.
- **Warning:** Do not cross the lines shown in the maze. These lines are invisible on the bomb.



On the Subject of Passwords

Fortunately this password doesn't seem to meet standard government security requirements: 22 characters, mixed case, numbers in random order without any palindromes above length 3.

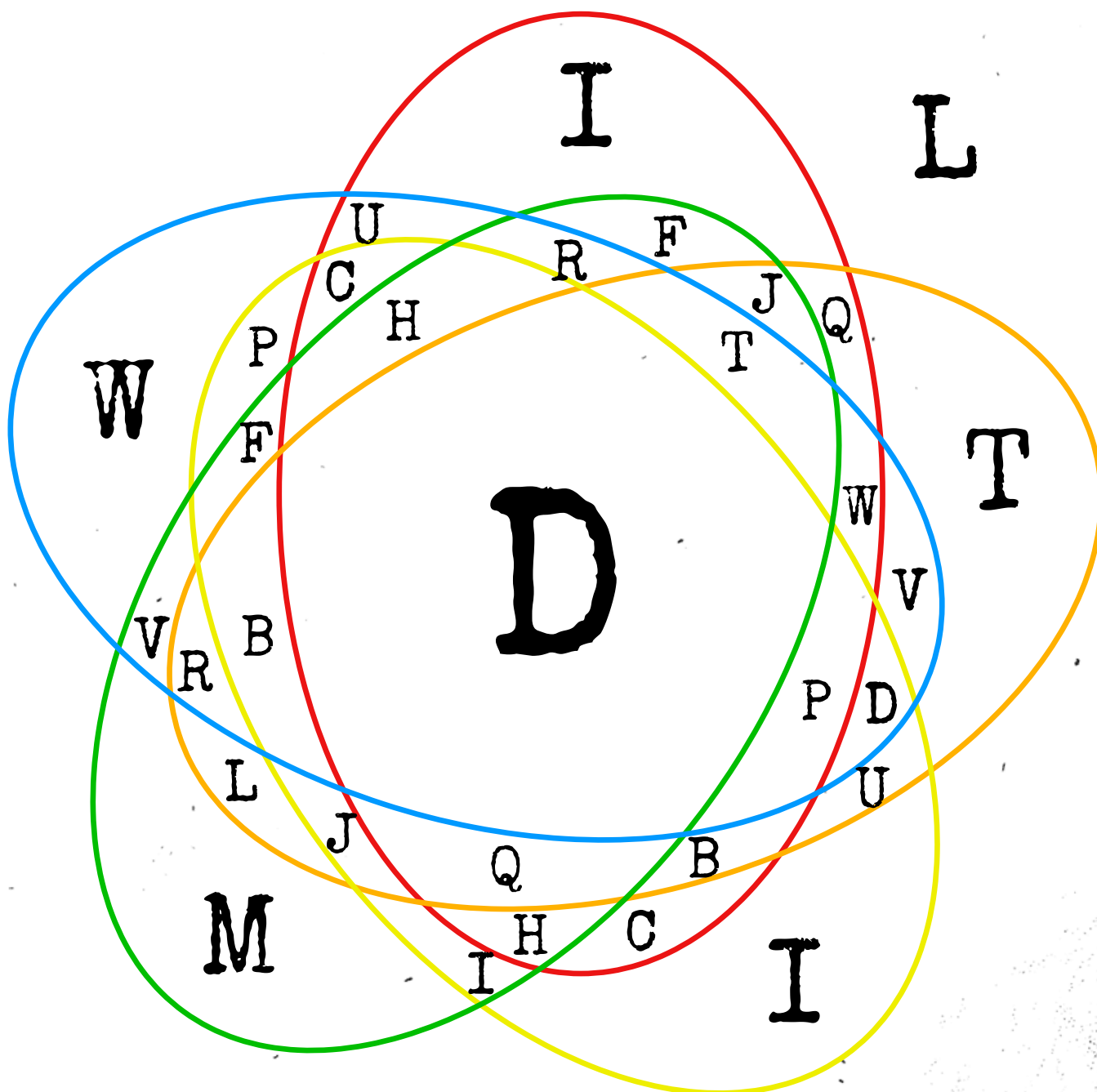
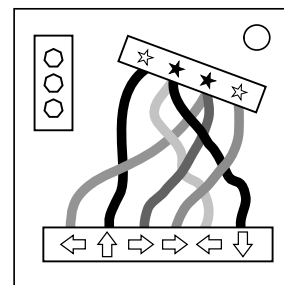
- The buttons above and below each letter will cycle through the possibilities for that position.
- Only one combination of the available letters will match a password below.
- Press the submit button once the correct word has been set.



about	after	again	below	could
every	first	found	great	house
large	learn	never	other	place
plant	point	right	small	sound
spell	still	study	their	there
these	thing	think	three	water
where	which	world	would	write

Complicated Wires just isn't complicated enough.

- Look at each wire: there is a “★” symbol above the wire and an arrow symbol below the wire, as well as three small LEDs on the side.
- For **each** wire, use the Venn diagram below to decide whether or not to cut the wire. The meanings of the colors and letters in the Venn diagram are described on the next page.



Red	Orange	Yellow	Green	Blue
The wire is red, yellow, blue, or white.	The wire shares the same color as its arrow.	The wire's star is black.	The wire's position on the bottom is even.	The wire crosses another wire. ^[1]

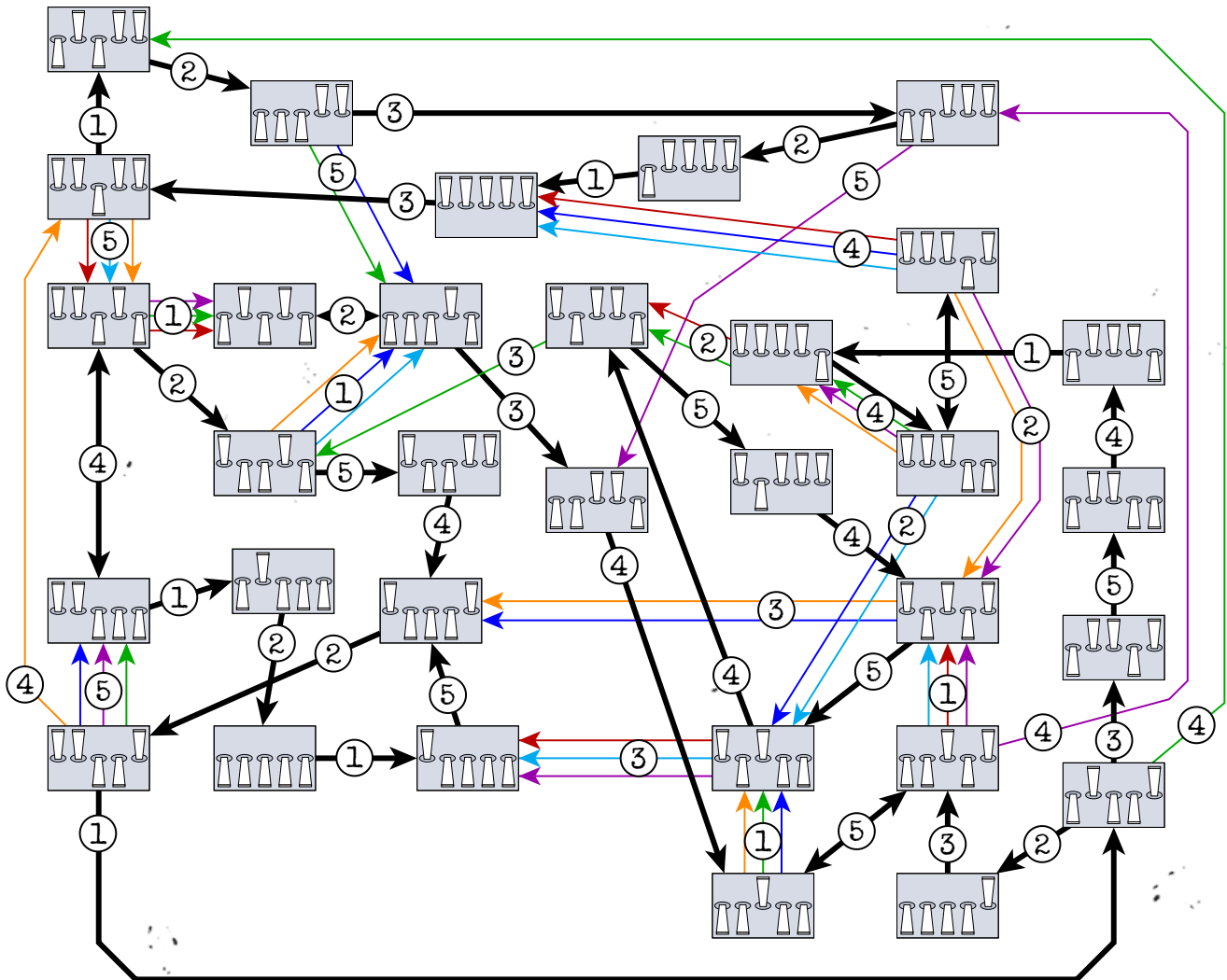
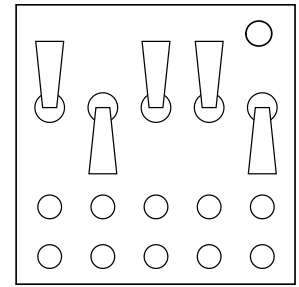
Letter	Instruction
C	Cut the wire.
F	Always cut the wire, but only cut it first.
L	Always cut the wire, but only cut it last.
W	Cut the wire if more of the LEDs are on than off.
T	Cut the wire if the top LED is on.
U	Cut the wire if its arrow points up or down.
M	Cut the wire if the arrow points down or right.
H	Cut the wire if the wire shares a star with another wire.
P	Cut the wire if its position at the bottom is equal to the number of ports.
B	Cut the wire if its position at the bottom is equal to the number of batteries.
I	Cut the wire if its position at the bottom is equal to the number of indicators.
Q	Cut the wire if the color of the wire is unique.
J	Cut the wire if, at the bottom, it is adjacent to an orange or purple wire.
V	Cut the wire if the serial number has a vowel, or if the bomb has a USB port.
R	Cut the wire if its arrow direction is unique.
D	Do not cut the wire.

[1] Specifically, two wires are considered “crossing” if their top connectors (near the stars) are in the opposite order from their bottom connectors (near the arrows).

On the Subject of Colored Switches

It's only five switches. How hard can it be?

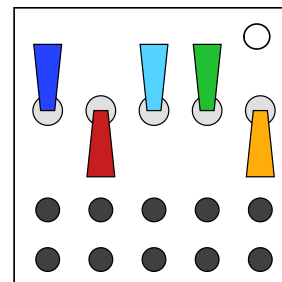
- The module has five switches on it. Each switch may be orange, green, purple, turquoise, blue, or red.
- The diagram below indicates which switch transitions are safe. A switch can be toggled only if it has the same color as the relevant arrow in the diagram. A black, thicker line indicates this switch can be toggled regardless of its color. The numbers indicate the switch being toggled (numbered from left to right).
- To defuse the module, first make any three permissible toggles. Then the LEDs below the switches come on. The switches must be brought into the position indicated by those LEDs.



Appendix: Identifying lookalikes of *Colored* Switches

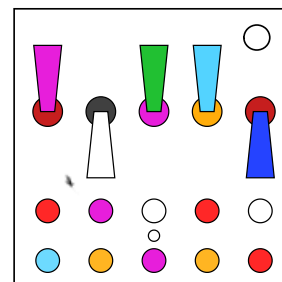
If there are no lit LEDs:

Colored Switches



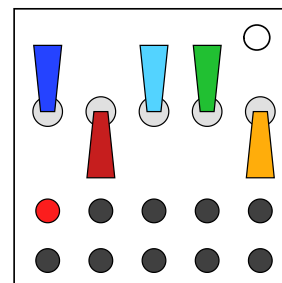
If the sockets are colored and there is a middle LED between the top and bottom row of LEDs:

Multicolored Switches



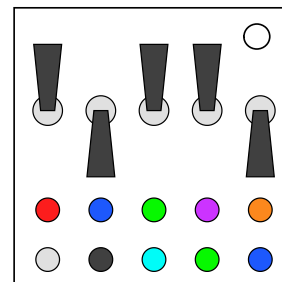
If the first LED is lit:

Recolored Switches



If all of the switches are black:

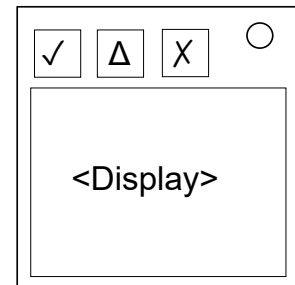
Uncolored Switches



On the Subject of Web Design

Welcome to web design class. The first rule of web design is: You do not talk about Comic Sans MS. The second rule is: You DO NOT talk about Comic Sans MS. The third rule is: The word "color" is NEVER spelled with a U.

- The module displays a snippet of a CSS file on a screen.
- Based on the information given, you have to Accept (✓), Consider (Δ), or Reject (X) the code.



Step 1: Select the website

Find the website this code was quoted from. Use the **Selector** part of the code (the part before { }) to narrow it down to one site, then note down its **Threshold** value.

<u>Edison Daily (news site)</u> Elements: body, a, h3, blockquote IDs: #header, #comments Classes: .post, .title, .author Threshold: #00FF00	<u>Buddymaker (social media)</u> Elements: div, span, img, a IDs: #msg, #cover, #content, #sidebar Classes: .post, .title, .share Threshold: #8040C0
<u>PNGdrop (image hosting)</u> Elements: div, img IDs: #main, #comments, #fullview Classes: .username, .share, .large Threshold: #BADA55	<u>BobIRS (chatroom)</u> Elements: ul, ol, img, b, i IDs: #sidebar Classes: .avatar, .username Threshold: #03E61E
<u>Vidhost (video hosting)</u> Elements: div, iframe, b, i IDs: #main, #rating, #comments Classes: .username, .share, .channel Threshold: #60061E	<u>Go Team Falcon online (online game)</u> Elements: body, iframe IDs: #rating, #comments Classes: .rating, .fullscreen Threshold: #501337
<u>Stufflocker (cloud storage)</u> Elements: div, h3, img, iframe IDs: #sidebar, #download Classes: .menu, .author Threshold: B020E5	<u>Steel Nexus (forum)</u> Elements: body, div, img, blockquote IDs: #header, #content, #sidebar Classes: .avatar, .reply Threshold: #BEA61E

Step 2: Find the color target

Find the first color name in the CSS file, then use the following table to determine a hexadecimal value of that color. This value is the **Color Target** that you will need later. If no color name is displayed, use #7F7F7F.

Color	Hex value	Color	Hex value	Color	Hex value
Blue	#0000FF	Yellow	#FFFF00	Red	#FF0000
Green	#00FF00	White	#FFFFFF	Orange	#FFA500
Purple	#800080	Magenta	#FF00FF	Gray	#808080

Note: The hexadecimal value of a color is denoted in #RRGGBB format.

Step 3: Calculate site score

Calculate the site score as follows:

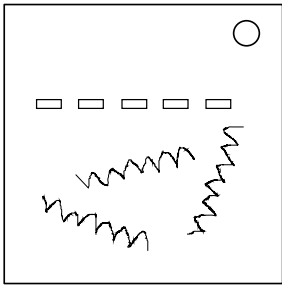
- Start with the number of lines of code inside the {curly braces}. Counting semi-colons is a good idea because it is a line terminator.
- +3 score each for:
 - R value of the Color Target is less than R value of the threshold.
 - G value of the Color Target is greater than or equal to G value of the threshold.
 - B value of the Color Target is greater than B value of the threshold.
- +2 score for each margin/padding.
- +1 score for each border/border-radius unless it's 0px or 50%.
- -1 score for each z-index without a position.
- +1 score for each font-family unless it's "Comic Sans MS", which gives a -5 score.
- +2 score for each box-shadow/text-shadow unless it's none.
- ×2 score if the buttons on the module are colored, or -3 score if they are gray.
- If your score is now negative or 0, keep adding 16 until it's positive.
- Keep adding up the digits of the number until you are left with a single digit. Then use the following table to determine which button to push.

Accept (✓)	Consider (Δ)	Reject (X)
2, 3, 5, 7	6, 8	1, 4, 9

On the Subject of Binary LEDs

Binary is a time-honored tradition of communication with tiny blinking lights.

- Interpret the binary code from the five LEDs. The bit farthest left is the most significant.
- The code will follow one of the eight numeric sequences below. The sequence oscillates forwards and backwards without repeating the ends.
- Some numbers in the sequence have a letter underneath it. Cut the wire of the matching color while this part of the sequence is displayed.
- Only one wire needs to be cut successfully.



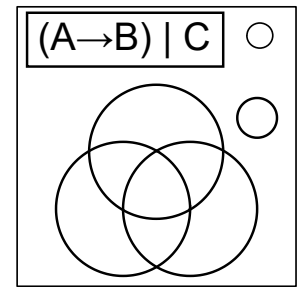
17	15	6	2	24	8	26	25	21	24	1	15	18	8
			G		R		B						
18	15	19	31	12	6	19	21	11	16	19	2	1	29
				B				G			R		
8	25	1	15	20	15	9	3	6	24	1	24	5	26
	G	B							R				
21	27	6	12	27	20	7	1	19	15	3	13	9	28
							B	R				G	
3	21	14	22	7	28	16	27	22	17	26	2	31	15
					G			B		R			
8	22	30	19	1	25	31	16	9	7	6	13	9	7
R						B				G			
5	18	12	7	5	12	31	16	10	15	17	9	12	25
		R			G				B				
4	20	18	25	20	4	24	29	17	16	12	16	29	19
					G				R	B			

Color Key:	R	red
	G	green
	B	blue

On the Subject of Boolean Venn Diagrams

Why is there a big Venn diagram? Why are there some weird symbols? Oh no...

- This module has eight buttons, one for each enclosed section of the Venn diagram and one representing the area not enclosed in any section of the diagram.
- The three circles are referred to as "A" (top), "B" (bottom left), and "C" (bottom right).
- For each section, use the boolean logic expression displayed above the Venn diagram to determine if that section is "true" or "false" by using the following rules:
 - If the section is enclosed in a circle, then the value for that variable is considered to be "true". Otherwise, it is "false".
Example: The middle section is enclosed in all three circles, so "A", "B", and "C" should all be considered to be "true" while evaluating that section.
 - Evaluate the operator inside the parentheses before the one outside of the parentheses.
 - Images describing each operator can be found below (gray regions represent "true").
 - If the section evaluates to "true", press the button that corresponds with it. The section will turn green.
- Solve the module by pressing the buttons corresponding to all of the "true" sections.
- Note: if an incorrect button is pressed, a strike will be issued and the section will turn red.

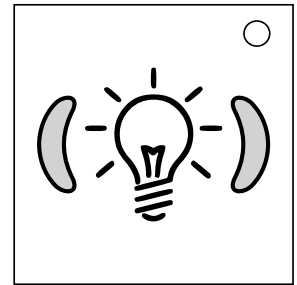


$X \wedge Y$ AND	$X \vee Y$ OR	$X \underline{\vee} Y$ XOR	$X \rightarrow Y$ IMPLIES
$X \mid Y$ NAND	$X \downarrow Y$ NOR	$X \leftrightarrow Y$ XNOR	$X \leftarrow Y$ IMPLIED BY

On the Subject of The Bulb

How many bomb defusal experts does it take to screw in a light bulb?

This module has two buttons labeled **I** and **O** and a light bulb, which is either see-through (translucent) or opaque, and is one of six colors: blue, green, purple, red, white or yellow.



If you incur a strike because you pushed a wrong button, ignore it and continue. If you incur a strike because you unscrewed or screwed in the bulb at an incorrect time, you must undo that before continuing.

Begin at Step 1 below.

While the bulb is screwed in, a long press on either button will reset the module back to Step 1. Attempting this while the bulb is out will incur a strike.

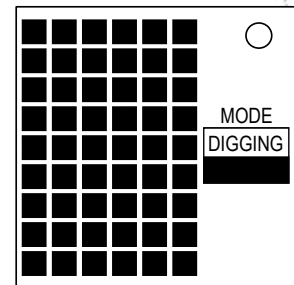
- Step 1**
- If the light is on and the bulb is see-through, press **I** and go to **Step 2**.
 - If the light is on and the bulb is opaque, press **O** and go to **Step 3**.
 - Otherwise, unscrew the bulb and go to **Step 4**.
- Step 2**
- If the bulb is red, press **I**, then unscrew it and go to **Step 5**.
 - If the bulb is white, press **O**, then unscrew it and go to **Step 6**.
 - Otherwise, unscrew the bulb and go to **Step 7**.
- Step 3**
- If the bulb is green, press **I**, then unscrew it and go to **Step 6**.
 - If the bulb is purple, press **O**, then unscrew it and go to **Step 5**.
 - Otherwise, unscrew the bulb and go to **Step 8**.
- Step 4**
- If the bomb has any of the following indicators: CAR, IND, MSA or SND, press **I** and go to **Step 9**.
 - Otherwise, press **O** and go to **Step 10**.
- Step 5**
- If the light went off at Step 1, press the same button again, then screw the bulb back in.
 - Otherwise, press the button you haven't yet pressed, then screw the bulb back in.
- Step 6**
- If the bulb went off when you pressed **I**, press the button that you pressed in Step 1, then screw the bulb back in.
 - Otherwise, press the button that you pressed in Step 2 or 3, then screw the bulb back in.

- Step 7**
- If the bulb is green, press **I**, remember SIG and go to **Step 11**.
 - If the bulb is purple, press **I**, then screw it back in and go to **Step 12**.
 - If the bulb is blue, press **O**, remember CLR and go to **Step 11**.
 - Otherwise, press **O**, then screw the bulb back in and go to **Step 13**.
- Step 8**
- If the bulb is white, press **I**, remember FRQ and go to **Step 11**.
 - If the bulb is red, press **I**, then screw it back in and go to **Step 13**.
 - If the bulb is yellow, press **O**, remember FRK and go to **Step 11**.
 - Otherwise, press **O**, then screw the bulb back in and go to **Step 12**.
- Step 9**
- If the bulb is blue, press **I** and go to **Step 14**.
 - If the bulb is green, press **I**, then screw it back in and go to **Step 12**.
 - If the bulb is yellow, press **O** and go to **Step 15**.
 - If the bulb is white, press **O**, then screw it back in and go to **Step 13**.
 - If the bulb is purple, screw it back in, then press **I** and go to **Step 12**.
 - Otherwise, screw the bulb back in, then press **O** and go to **Step 13**.
- Step 10**
- If the bulb is purple, press **I** and go to **Step 14**.
 - If the bulb is red, press **I**, then screw it back in and go to **Step 13**.
 - If the bulb is blue, press **O** and go to **Step 15**.
 - If the bulb is yellow, press **O**, then screw it back in and go to **Step 12**.
 - If the bulb is green, screw it back in, then press **I** and go to **Step 13**.
 - Otherwise, screw the bulb back in, then press **O** and go to **Step 12**.
- Step 11**
- If the bomb has the remembered indicator, press **I**, then screw the bulb back in.
 - Otherwise, press **O**, then screw the bulb back in.
- Step 12**
- If the light is now on, press **I**.
 - Otherwise, press **O**.
- Step 13**
- If the light is now on, press **O**.
 - Otherwise, press **I**.
- Step 14**
- If the bulb is opaque, press **I**, then screw the bulb back in.
 - Otherwise, press **O**, then screw the bulb back in.
- Step 15**
- If the bulb is see-through, press **I**, then screw the bulb back in.
 - Otherwise, press **O**, then screw the bulb back in.

On the Subject of Minesweeper

Somehow Minesweeper on a bomb feels oddly redundant.

- A Minesweeper module has a grid of cells with a mode toggle button.
- By pressing the toggle button you can toggle between two different modes.
 - Dig Mode - Which uncovers a cell that you think doesn't have a mine.
 - Flag Mode - Which flags a cell that you think has a mine.



Locating the starting cell

The first cell to dig will be one of the colored squares. Start with the second digit in the serial number. If it is zero, substitute 10. Count that many colored squares in reading order, wrapping around to the top left of the grid if necessary.

Use the color found to look up a value in the table. Add this number to the numeric value of the first letter in the serial number, with A = 1, B = 2, etc. Starting from the bottom right corner, count that many colored squares in reverse reading order, looping back if necessary. This gives you the first location to dig.

Color	#
Red	5
Orange	2
Yellow	3
Green	1
Blue	6
Purple	4

NOTE: You cannot dig other cells until you have dug the correct starting cell and digging an incorrect starting cell will give you a strike, regardless of whether it's a mine or not.

Solving Minesweeper

Each number on the grid represents how many mines there are in the surrounding 8 cells. Use these numbers to figure out where each mine is. Once you have achieved either all non-mine cells dug or all mine filled cells flagged, the module will solve. Digging up a cell with a mine under it will expose the mine and a strike will be given.

NOTE: Clicking on a number while digging will dig all cells around it that aren't flagged. Holding down on a cell will flag it instead of digging it.

On the Subject of Cruel Piano Keys

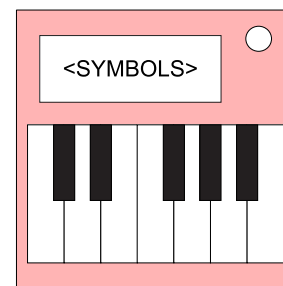
The devil's interval approaches...

See Appendix A for indicator identification reference.

See Appendix B for battery identification reference.

See Appendix C for port identification reference.

See the third page for serialism & music terminology reference.



- A cruel piano keys module will present with 4 musical symbols in the top indicator and a 12-note keyboard to input with.
- Each rule consists of one or more required symbol(s) and optional further requirements based on the bomb casing.
- Follow the list of rules down in **Table 2** until one matches the criteria for the module and bomb.
- Then use the lookup criteria to find the prime 12-tone row from **Table 1**.
- Then apply the according transformation from **Table 2** to the 12-tone row, and execute this final sequence.
- A failed attempt will require re-entry of the entire note sequence.

Table 1.

#	<u>Prime 12-tone Sequence</u>	#	<u>Prime 12-tone Sequence</u>
0	F D F# G# C B A# C# G E D# A	5	C D# F# D F C# B A G A# E G#
1	A# A C E C# D D# G B F# G# F	6	G# C A# C# E G B D# A D F F#
2	F# B A G# D C G C# F D# E A#	7	E A C# B G G# A# D# F# F C D
3	E D# D F# F A# G# C# C B G A	8	G# D# D E A# C# F# G F A C B
4	D E A A# C B C# G# F F# D# G	9	D# G# C B D C# F# A# F G A E

Table 2.

<u>Required Symbol(s)</u>	<u>Further Requirements</u>	<u>Lookup Index</u>	<u>Transformation</u>
and ~	2 or more indicators (lit or unlit)	Left-most digit in serial number	RI
# or ×	An empty port plate	Number of battery holders*	P, transpose down by 'x' semitones, where 'x' = number of minutes remaining
⌒ or □	2 or more of a certain type of port	Least significant digit of number of completed modules	I
and ∕	2 or more port plates	9 minus the number of unlit indicators†	R
¢ or C	Serial contains 1 or more vowels	Least significant digit of number of strikes	R, transpose down by 3 semitones
‡ or ~	Even number of batteries	DVI-D present: 7 Otherwise: 3	P, transpose up by 'x' semitones, where 'x' = number of ports‡
♭ or {	An indicator with no vowels in the label	8	I
□ or ∕	Less than 2 ports†	4	R
or ×	(No other requirements)	5	P
If none of these rules apply, revert back to the <u>Normal</u> Piano Keys ruleset and play the given note sequence normally.			

Notes:

*: If the number of battery holders exceeds 9, continually subtract 10 until you have a result in the 0 to 9 range (inclusive).

†: If the result is negative, continually add 10 until you have a result in the 0 to 9 range (inclusive).

‡: The Stereo RCA port does not count as 2 separate ports; the Red & White connectors are part of the same singular port.

Serialism & Music Terminology

To clarify, the note below a C would be a B, and similarly, the note after a B would be a C. The 12 tones on the piano essentially wrap around.

The Prime sequence (or 'P' for short), is the original or base form of the 12-tone row. No transformation takes place.

The Retrograde sequence (or 'R' for short), takes the Prime sequence, but executes it in reverse order. For example, the Retrograde of the Prime row A B C D E would be E D C B A.

The Inverse sequence (or 'I' for short), takes the Prime sequence, but the intervals between the notes are inverted. For example, take the interval from A to B; the interval is +2 semitones, as it takes you 2 semitones to get from A to B (A goes to A[#] then B). The inversion of this interval would be -2 semitones. Therefore, the inverted sequence would be A then G, as G is -2 semitones away from A (A goes to G[#] then G).

As an extended example, the Inversion of the Prime row A B C D E would be A G F[#] E D; the first note always remains the same, and all the other notes get inverted relative to that note.

The Retrograde Inverse sequence (or 'RI' for short), takes the Inverse sequence in Retrograde. For example, the Retrograde Inverse of the Prime row A B C D E would take the Inverse first (which is A G F[#] E D), and then the Retrograde of this Inverse would be D E F[#] G A.

Transpositions apply a translation of the tone row up or down by a given number of semitones. For example, the Prime row A B C D E transposed up by 1 semitone would be A[#] C C[#] D[#] F.

An Interval is the tonal distance between two distinct notes and is usually measured in semitones. For example, the interval from G to B is up 4 semitones.

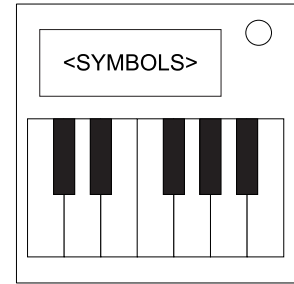
On the Subject of Festive Piano Keys

Let's all gather round for a family sing-a-long!

See Appendix A for indicator identification reference.

See Appendix B for battery identification reference.

See Appendix C for port identification reference.



- A festive piano keys module will present with 3 musical symbols in the top indicator and a 12-note keyboard to input with; This will be different to the regular piano keys module by a different backing color.
- Each rule consists of one or more required symbol(s) and optional further requirements based on the bomb casing.
- Follow the list of rules down until one matches the criteria for the module; then execute the sequence of notes listed.
- A failed attempt will require re-entry of the entire note sequence.

<u>Required Symbol(s)</u>	<u>Further Requirements</u>	<u>Note Sequence</u>
//	More even digits in the serial number than odd	E ^b F E ^b C A ^b F E ^b
§ or ♪	Any duplicate characters or digits in serial number	C [#] B A F [#] G [#] A G [#] F [#]
↯ and *	(No other requirements)	G A G E G A G E
∨ or ▯	At most 2 types of port	E ^b E ^b D ^b A ^b E ^b E ^b F D ^b
^	Lit indicator with vowel	B A G E ^b D A B A G
♫ or ♪	3 or more AA batteries	F [#] G A A D B A G E D
○ and ○	(No other requirements)	G E F G C B C D C B A G
> or ^ or ∨	Serial number contains 1 or 9	G G G G G G G B ^b E ^b F G
§ or § or //	(No other requirements)	D D D C [#] C [#] C [#] B C [#] B F [#]
(No requirement)	(No other requirements)	B ^b A B ^b G (Play sequence <u>x</u> times, where <u>x</u> equals largest digit in serial + 1)

On the Subject of Piano Keys

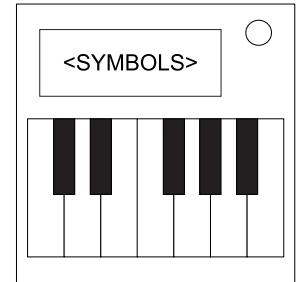
What do you get when you drop a piano down a mine shaft? A flat minor.

See Appendix A for indicator identification reference.

See Appendix B for battery identification reference.

See Appendix C for port identification reference.

See the next page for piano/keyboard reference.

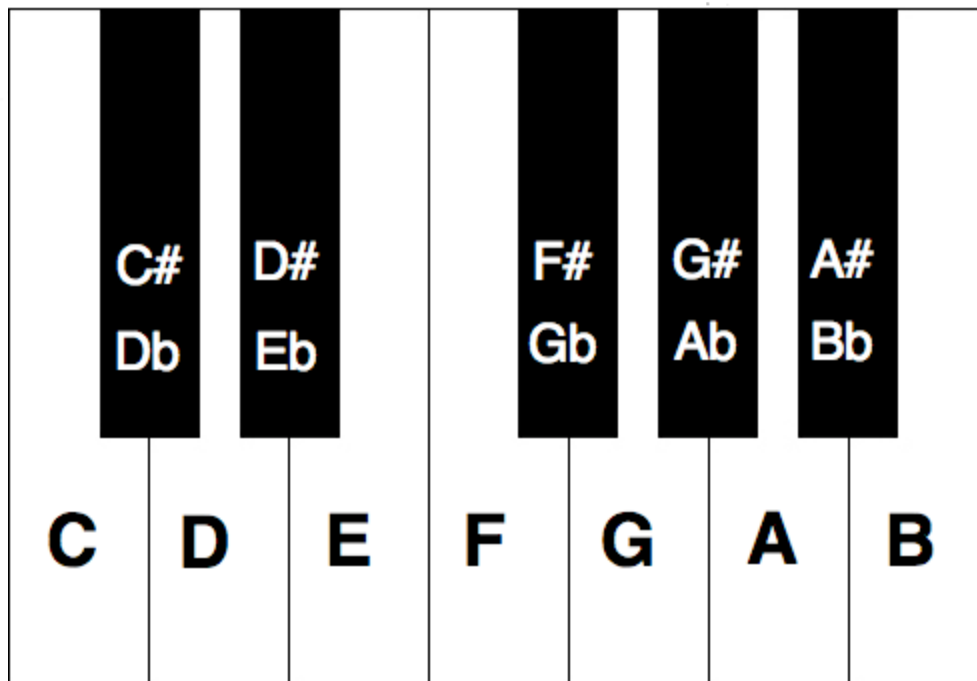


- A piano keys module will present with 3 musical symbols in the top indicator and a 12-note keyboard to input with.
- Each rule consists of one or more required symbol(s) and optional further requirements based on the bomb casing.
- Follow the list of rules down until one matches the criteria for the module; then execute the sequence of notes listed.
- A failed attempt will require re-entry of the entire note sequence.

<u>Required Symbol(s)</u>	<u>Further Requirements</u>	<u>Note Sequence</u>
$\flat$	Last digit of serial number is even	$B\flat B\flat B\flat B\flat G\flat A\flat B\flat A\flat B\flat$
C or $\sharp$	2 or more battery holders	$E\flat E\flat D D E\flat E\flat D E\flat E\flat D D E\flat$
$\natural$ and $\ominus$	(No other requirements)	$E F\sharp F\sharp F\sharp F\sharp E E E$
C or $\sim$	RCA port is present	$B\flat A B\flat F E\flat B\flat A B\flat F E\flat$
B	SND indicator is present and lit	$E E E C E G G$
w or $\ominus$ or C	3 or more batteries	$C\sharp D E F C\sharp D E F B\flat A$
$\flat$ and $\sharp$	(No other requirements)	$G G C G G C G C$
C or w	Serial number contains a 3, 7 or 8	$A E F G F E D D F A$
$\natural$ or $\sim$ or B	(No other requirements)	$G G G E\flat B\flat G E\flat B\flat G$
(No requirement)	(No other requirements)	$B D A G A B D A$

Piano/Keyboard Reference

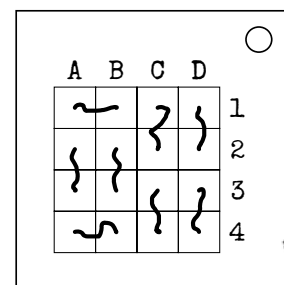
Use the following graphic as a reference to how tones are mapped onto a standard 12-note piano/keyboard.



On the Subject of Wire Placement

Sometimes, the wire may look like a face, calming you down... and then you explode.

- This module contains a grid of wires.
- There are always 8 wires on it.
- Wires can be red, blue, yellow, black or white.
- In the following table, use only the column corresponding to the color of the wire connected to C3.
- Cut a wire if it is a specific color and is connected to a specific spot on the grid as indicated in the table.

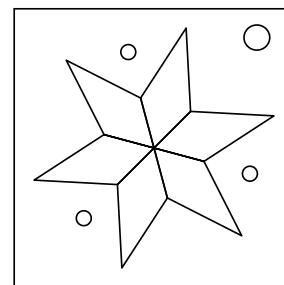


	Wire connected to C3 is				
	Black	Blue	Red	White	Yellow
Cut if color	is connected to:				
Yellow	D2	D1	D2	A2	D1
Blue	A2	C4	A1	C4	D4
White	D3	D2	D4	B3	B2
White	B2	C1	B4	A1	C1
Red	A1	B3	C4	B2	B3
Blue	C3	C2	C1	D3	B1
Black	B1	D4	A4	D2	B4
Red	C4	D3	B1	C1	C2
Yellow	A3	C3	A2	A4	A3
Yellow	D1	A1	B2	B4	A4

On the Subject of Simon Screams

He's angry! He's furious! He's enraged! He's had it!

- This module has six lights colored red, orange, yellow, green, blue and purple. These will flash in a sequence that grows longer with each stage. There are 3 stages.
- At each stage, consider the whole sequence of flashes. In the large table, find the first applicable row and the correct column. From that entry, take the letter in the position corresponding to the current stage and look at its corresponding column in the smaller table.
- Go through that column from top to bottom and press every color whose condition applies. The colors are (R)ed, (O)range, (Y)ellow, (G)reen, (B)lue, (P)urple, and “#” means “serial number”.
- Every time the sequence flashes again, your current stage's input is reset.



	A	C	D	E	F	H
≥ 3 indicators	Y	O	G	R	B	P
≥ 3 ports	P	Y	R	B	O	G
≥ 3 digits in #	O	G	B	P	R	Y
≥ 3 letters in #	G	B	O	Y	P	R
≥ 3 batteries	R	P	Y	O	G	B
≥ 3 bat. holders	B	R	P	G	Y	O

Stage 1: first flashing color

Stage 2: second flashing color

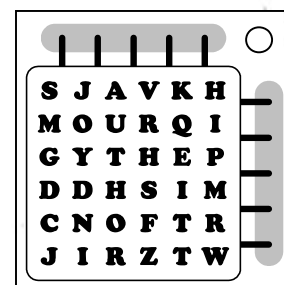
Stage 3: third flashing color

	red	orange	yellow	green	blue	purple
If three adjacent colors flashed in clockwise order	FFC	CEH	HAF	ECD	DDE	AHA
Otherwise, if a color flashed, then an adjacent color, then the first again	AHF	DFC	ECH	CDE	FEA	HAD
Otherwise, if at most one color flashed out of red, yellow, and blue	DED	ECF	FHE	HAA	AFH	CDC
Otherwise, if there are two colors opposite each other that didn't flash	HCE	ADA	CFD	DHH	EAC	FEF
Otherwise, if two adjacent colors flashed in clockwise order	CAH	FHD	DDA	AEC	HCF	EFE
Otherwise	EDA	HAE	AEC	FFF	CHD	DCH

Oдноśnie Wykreślanki

LZIEAJDTRUDNIEJSZEBNCOJWNIŻPQIEYBZSIEWYDAJEYWH

Na module znajduje się pole z 36 literami. Niektóre z tych liter tworzą słowa, które mogą być odczytane od tyłu i pojawić się w dowolnym kierunku.



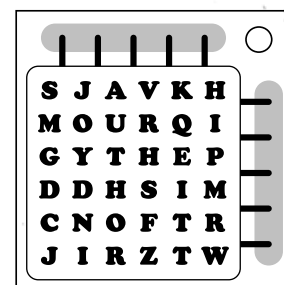
1. Poniższa siatka zawiera kwadraty z literami w narożnikach. Dla każdej z czterech liter w rogach wyświetlacza, znajdź kwadrat na siatce, które zawiera tę literę w tym samym narożniku.
2. Gdy już wszystkie odpowiednie kwadraty zostaną odnalezione, użyj ostatniej cyfry numeru seryjnego (parzystej lub nieparzystej), aby zdecydować których słów należy szukać.
3. Tylko jedno z tych słów pojawi się na wyświetlaczu. Zaznacz pierwszą i ostatnią literę poprawnego słowa, aby unieszkodliwić moduł. Bomba odnotuje pomyłkę, jeśli zaznaczone zostanie jakiekolwiek inne słowo.

[parzyste]	CHEŁM	JERZYK	ŁĄD	MAGNES	BATUTA	
—	—	—	—	—	—	
[nieparz.]	CHOĆ	DRÓŻKA	WRZÓD	ROMB	GRAD	
		V	U	S	Z	
BÓG	BOOM	HART	LITERA	JERZY	DRUŻKA	SMOG
—	—	—	—	—	—	—
MORZE	TRUDNE	CZAT	BRÓD	CHODŹ	ALARM	LUTY
	P	Q	N	X	F	Y
HUK	AGENT	MAGNEZ	BUK	KABOOM	WRZUT	PRÓG
—	—	—	—	—	—	—
MIEDŹ	BATUT	GRAT	JĘZYK	MINI	KOD	CHART
	T	I	M	E	D	A
TRUDNO	STÓG	LUT	AJENT	BOMBA	PUK	TEST
—	—	—	—	—	—	—
HEŁM	PUSTE	SMOK	BUM	ATUT	JĘŻYK	MAG
	K	B	W	H	J	O
LONT	CZAD	DALEJ	MIEĆ	BUG	LÓD	KURDE
—	—	—	—	—	—	—
SŁOWO	MÓW	SERIAL	STUK	METR	MOŻE	BRUD
		R	L	C	G	
	RZUT	LITR	NIC	RĄB	KOS	
	—	—	—	—	—	
	MILI	KOT	COFNIJ	LUD	MAK	

On the Subject of Word Search

LZIEAJDHARDERBNCOJWTHANPQIEYBZITLOOKSYWH

A field of 36 letters will appear on the screen within the module. Some of these letters will spell out words, which may be spelled backwards and appear in any direction.



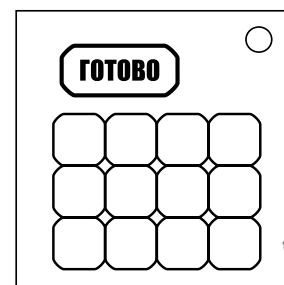
1. The chart below contains boxes with letters in the corners. For each of the four letters in the corners of the display, find a box on the chart that has that letter in the same corner.
2. Once all the relevant boxes have been located, use the last digit of the serial number (even or odd) to determine the correct words to reference.
3. Only one of those words will appear on the display. Select the first and last letter of the correct word to disarm the module. The bomb will record a strike if any other words are selected.

[even] —	HOTEL —	SEARCH —	ADD —	SIERRA —	FINISH —	
[odd] —	DONE —	QUEBEC —	CHECK —	FIND —	EAST —	
	V		U		S	Z
PORT —	BOOM —	LINE —	KABOOM —	PANIC —	MANUAL —	DECOY —
COLOR —	SUBMIT —	BLUE —	ECHO —	FALSE —	ALARM —	CALL —
P	Q	N	X	F	Y	
SEE —	INDIA —	NUMBER —	ZULU —	VICTOR —	DELTA —	HELP —
TWENTY —	NORTH —	LOOK —	GREEN —	XRAY —	YES —	LOCATE —
T	I	M	E	D	A	
ROMEO —	TRUE —	MIKE —	FOUND —	BOMBS —	WORK —	TEST —
BEEP —	EXPERT —	EDGE —	RED —	WORD —	UNIQUE —	JINX —
K	B	W	H	J	O	
GOLF —	TALK —	BRAVO —	SEVEN —	MODULE —	LIST —	YANKEE —
LETTER —	SIX —	SERIAL —	TIMER —	SPELL —	TANGO —	SOLVE —
	R	L	C	G		
	CHART —	MATH —	READ —	LIMA —	COUNT —	
	OSCAR —	NEXT —	LISTEN —	FOUR —	OFFICE —	

Смежные буквы (Русский)

ВЕРХ, низ, лево, право, ВЕРХ, низ, лево, право...!

Убедитесь, что на кнопке сверху модуля написано "ГОТОВО", а не "SUBMIT". В противном случае, это неправильная инструкция.



- На каждой клавише написана буква.
- С помощью таблицы определите, какие клавиши нажать.
- Каждая из букв в колонке "Буква", граничащая слева или справа с одной из букв из колонки "Слева/Справа", должна быть нажата.
- Каждая из букв в колонке "Буква", граничащая сверху или снизу с одной из букв из колонки "Сверху/Снизу", должна быть нажата.
- Нажмите кнопку "ГОТОВО", чтобы подтвердить ответ. **Важно:** Если нужно нажать несколько клавиш, их можно нажимать в **любом** порядке.

Буква	Слева/Справа	Сверху/Снизу
А	БГМНТ	ДМТТШ
Б	АЦЬЮЯ	ЖТХЩЭ
В	ИЫЭЮЯ	БГЬЭЮ
Г	ВДЕИК	ЕЖИКР
Д	ВЙХЩЬ	АИМОХ
Е	АДЛУЬ	АЙМНТ
Ж	АДЙТУ	ЦЩЬЮЯ
З	ВМРЧЫ	БОУЩЬ
И	ЖКРФХ	БРЬЫЬ
Й	НПСЬЭ	ВДСХЩ
К	ЕЖЛУФ	ДЖМОЦ
Л	ЖЙШЬЬ	ЕЖЭЮЯ
М	ОФХЧШ	НСЦЧЫ
Н	ЙЛОФЬ	ЛУЦЩЬ
О	ГДЖЬЭ	ГЖЭКР
П	ВОРЦЫ	ГДНРУ

Буква	Слева/Справа	Сверху/Снизу
Р	БЭКЬ	ВГЙНЦ
С	БВЗЛЦ	БГЕЗИ
Т	ЕИМСЯ	ВЕЙЛФ
У	ИПЦЩЬ	АКЛПШ
Ф	АОПЦШ	АВДЗО
Х	ГЭСФЧ	ВЛТФЭ
Ц	КЛОХЩ	ЛПСХЬ
Ч	ЕРЩЭЮ	БНПРУ
Ш	КМСУЧ	ОЧЬЮЯ
Щ	ГИМТХ	АБИЙП
Ь	БДЖЙН	МФЧШЯ
Ы	БПЩЮЯ	ЭКСХЧ
Ь	ЕУЭЮЯ	ИЙСТУ
Э	АГЭПТ	ФШЫЮЯ
Ю	НРЧШЬ	ЭФШЫЭ
Я	ЭНТШЫ	КЧЬЫЬ

On the Subject of Adjacent Letters

UP, down, left, right, UP, down, left, right...!

- Every button will have a different letter on it.
- Check the table to determine which buttons to press.
- Each letter in the “letter” column that has one of the letters in the “left/right” column directly left or right of it on the keypad must be pressed.
- Each letter in the “letter” column that has one of the letters in the “above/below” column directly above or below it on the keypad must be pressed.
- Press Submit to submit your answer. **Note:** Multiple buttons can be pressed and they can be in **any** order.

Letter	Left/right	Above/below
A	GJMOY	HKPRW
B	IKLRT	CDFYZ
C	BHIJW	DEMTU
D	IKOPQ	CJTUW
E	ACGIJ	KSUWZ
F	CERVY	AGJPQ
G	ACFNS	HOQYZ
H	LRTUX	DKMPS
I	DLOWZ	EFNUV
J	BQTUW	EHIOS
K	AFPXY	DIORZ
L	GKPTZ	ABRVX
M	EILQT	BFPWX

Letter	Left/right	Above/below
N	PQRSV	AFGHL
O	HJLUZ	IQSTX
P	DMNOX	CFHKR
Q	CEOPV	BDIKN
R	AEGSU	BNOXY
S	ABEKQ	GMVYZ
T	GVXYZ	CJLSU
U	FMVXZ	BILNY
V	DHMNW	AEJQX
W	DFHMN	GLQRT
X	BDFKW	AJNOV
Y	BCHSU	EGMTW
Z	JNRSY	CLMPV

On the Subject of Caesar Cipher

Communication was dangerous back in the days. Can you figure out what the original message was?

Decipher the characters on the display with the help of Caesar's tactics. There's no delete button, so press those buttons carefully!

[DISPLAY] ○

A	B	C	D
E	F	G	H
I	J	K	L

For example: if the offset is **-2**, **D** becomes **B** and **L** becomes **J**. The table below can be used to calculate the required offset. If the offset column contains an **=** sign, that value should be used, regardless of other rules that apply.

Condition	Offset
Serial number contains a vowel	-1
Number of batteries	+1 per battery
The last digit of the serial number is even	+1
Indicator with label CAR is present	+1
Parallel port and lit indicator with label NSA is present	=0

On the Subject of English Tests

You've lived all your life writing however you wanted without giving a second thought to who would be reading what you misspelled. But now, your life depends on your grammar and orthography, and this bomb is very nitpicky. You should of paid more attention in you're English class.

...Oh, carp.

See Appendix: Grammar for more information.

- An English sentence with one *italic* word or phrase will be displayed on the large LCD display.
- Your goal is to select the correct word that fills in the blank.
- If multiple words appear to complete the sentence correctly, remember that this module is a pedantic prescriptivist!
- There are three rounds. Correctly complete all the sentences to disarm the module.
- If a mistake is made during the course of the test, the question number will reset to 1.

Question 1/3 ☐

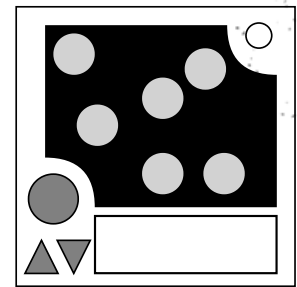
We need to *diffuse* this bomb quickly.

defuse

Magic

On the Subject of Friendship*Hey. I could defuse this bomb in ten seconds flat.*

- Locate the friendship symbols from the display in the following table.
- Out of the symbols indicating columns, disregard the one furthest left on the display that isn't exactly above or below any other friendship symbol on the display.
- Out of the symbols indicating rows, disregard the one furthest up on the display that isn't on the same height as any other friendship symbol on the display.
- Select one of the Elements of Harmony located at the intersections of the remaining rows and columns.



	J	G	U	K	V	8	L	C	H	4	W	P	M	R	
	7	S	8	U	N	J	9	Y	F	P	Q	C	R	4	
	Q	R	H	4	F	7	J	E	8	T	N	9	A	X	
	D	3	S	H	U	E	T	P	V	J	L	A	4	7	
	A	F	3	T	M	P	R	W	S	X	U	N	G	B	
	V	K	G	P	Q	D	U	L	3	H	M	R	E	C	
	4	9	T	F	B	X	D	U	Y	3	R	L	H	M	
	G	4	9	J	8	3	X	K	A	Y	S	W	7	D	
	K	T	F	B	J	Q	3	S	E	C	P	U	W	L	
	S	M	A	C	7	H	E	B	G	F	V	X	L	N	
	8	7	V	L	9	R	K	D	T	Q	B	Y	X	A	
	W	8	4	Q	G	Y	V	T	7	N	3	B	C	P	
	M	A	W	9	H	K	Y	J	N	D	X	E	8	F	
	Y	N	B	G	W	S	M	Q	K	9	C	V	D	E	

A = Altruism

B = Amicability

C = Authenticity

D = Benevolence

E = Caring

F = Charitableness

G = Compassion

H = Conscientiousness

J = Consideration

K = Courage

L = Fairness

M = Flexibility

N = Generosity

P = Helpfulness

Q = Honesty

R = Inspiration

S = Kindness

T = Laughter

U = Loyalty

V = Open-mindedness

W = Patience

X = Resoluteness

Y = Selflessness

3 = Sincerity

4 = Solidarity

7 = Support

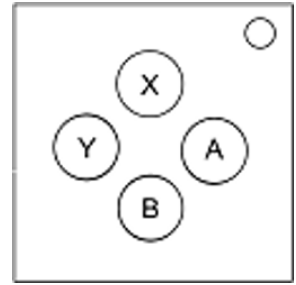
8 = Sympathy

9 = Thoughtfulness

On the Subject of Press X

Due to the fact that gamepads and controllers exist, we unfortunately had to settle with "Press X".

SEE APPENDIX PX100 FOR A LIST OF PRIME NUMBERS UNDER 100.



- This module consists of a four buttons, with the labels "X", "Y", "A" and "B".
- Using edge work, determine which button to press, then determine what time to press it.
- If a time rule states which button to press, it overrules the button rules.

Rules:

Which button to press:

Number of solved modules modulo 4↓	If there are more unlit indicators than lit indicators	If there are more lit indicators than unlit indicators	If there is an equal ammount of lit and unlit indicators*
0	A	B	Y
1	X	Y	A
2	B	A	X
3	Y	X	B

*0 lit idicators and 0 unlit indicators are equal amounts of both.

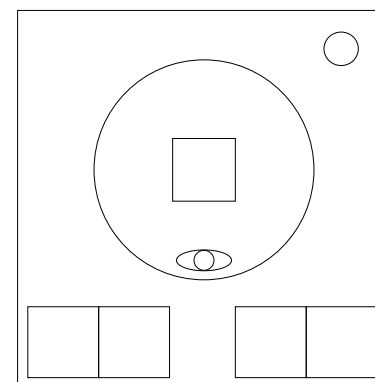
Time to press:

- If there is a lit indicator with label "CAR", the button you need to press is the "X" button and there are less than 2 batteries, press any button at any point in time.
- Otherwise, if there are 3+ batteries, press the button when the last digit in the countdown timer equal to the first digit in the serial number.
- Otherwise, if the button you need to press is the "A" button and there is either a 2 or 5 in the serial number, press "A" when the seconds digits in the countdown timer says 05 or 30.
- Otherwise, if the button you need to press is not "Y" and there is a lit indicator with the label "NSA", press the button when the seconds digit in the countdown timer are equal to each other.
- Otherwise, press the button when the seconds digits in the countdown timer add to 9.

On the Subject of Orientation

If the bomb doesn't kill us a brain haemorrhage will.

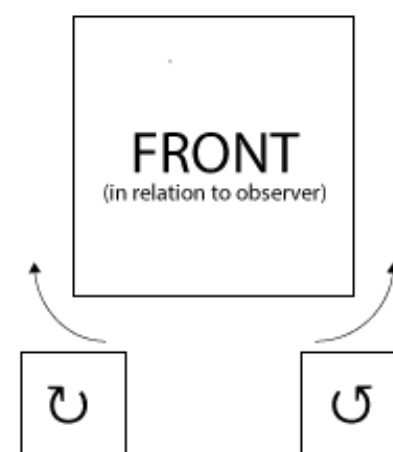
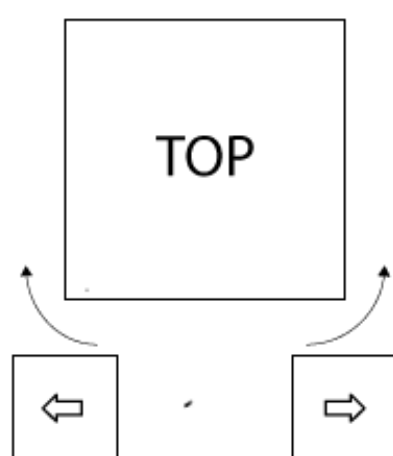
In order to diffuse this part of the bomb you will need good 3D orientation skills. A virtual cube needs to be rotated into a specific orientation using the four keys along the bottom. Unfortunately there is no display to indicate the current orientation of the virtual cube so you will have to imagine the state of the cube yourself.



The two keys in the bottom left will yaw the cube clockwise or anti-clockwise, respective to looking at the cube from the top.

The two keys in the bottom right will roll the cube clockwise or anti-clockwise, respective to the virtual observer. The virtual observer's position is indicated on the module as an eye. NOTE: The virtual observer's position may change.

For example, if the eye is at the bottom then it is facing the 'FRONT' face. Pressing 'Roll clockwise' will place the 'LEFT' face where the 'TOP' face is.



If the serial number on the bomb contains the letter R:

Rotate the cube so that the initial left face is in the same position as the initial top face, then press the SET button.

Otherwise, if the bomb has a lit indicator with the label TRN or has it has a lit/unlit indicator with the label CAR:

Rotate the cube so that the initial bottom face is in the same position as the initial right face, then press the SET button.

Otherwise, if the bomb has a PS2 port or there have been one or more strikes:

Rotate the cube so that the initial bottom face is in the same position as the initial front face and the initial left face is in the same position as the initial bottom face, then press the SET button.

Otherwise, if the serial number on the bomb contains either the number 7 or 8:

Rotate the cube so that the initial right face is in the same position as the initial bottom face and the initial back face is in the same position as the initial front face, then press the SET button.

Otherwise, if there are more than two batteries on the bomb or the virtual observer's initial position is facing the initial left face:

Rotate the cube so that the initial top face is in the same position as the initial bottom face, then press the SET button.

Otherwise:

Rotate the cube so that the initial top face is in the same position as the initial left face, then press the SET button.

On a strike:

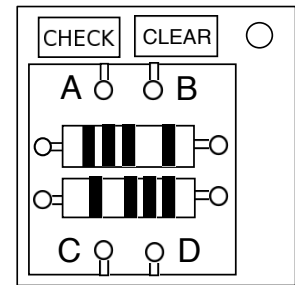
If you get strike then the virtual cube will be reset to the initial position, be aware you may need to select a new rule if the obersever is now in a different position.

On the Subject of Resistors

"It is easier to resist at the beginning than at the end."

– Leonardo da Vinci, on procrastination

The module contains 2 input pins (**A** and **B**), 2 resistors, and 2 output pins (**C** and **D**). Follow the rules to make the correct connections. To make a connection, click one pin and then another. Press **CLEAR** to remove all connections.



1. Take the first digit of the bomb's serial number (or 0 if there are no digits).
The *primary input* is **A** if even, **B** if odd.
2. Take the last digit of the bomb's serial number (or 0 if there are no digits).
The *primary output* is **C** if even, **D** if odd.
3. The *target resistance* in Ω is calculated as follows:
 1. Take the first two digits of the bomb's serial number.
e.g. 2E7XL9 $\rightarrow$ 27, ZJ3MLN $\rightarrow$ 3, ABCDEF $\rightarrow$ 0
 2. For each battery present on the bomb (up to a max of 6), multiply by 10.

4. Connect the primary input to the primary output, with the target resistance.

Note: all resistance values are checked to be within 5% accuracy.

5. If a lit **FRK** indicator is present, also connect the primary input to the other (secondary) output, with the target resistance.

Note: this means C and D will also be connected with some non-infinite resistance. This value is not checked as part of your solution, and so can be anything.

6. If step 5 did not apply and at least 1 *D cell battery* is present, connect the secondary input to the secondary output, with 0Ω resistance.

7. Press **CHECK** when finished to check the solution. All input/output pairs not mentioned should be disconnected.

Consult the following page to learn how to produce the target resistance.

Producing resistance

An input and output can be connected via one of five paths.

1. **No resistors**, 0Ω of resistance.

2. **Top resistor**.

3. **Bottom resistor**.

4. **Both resistors in serial**.

i.e. input → top resistor → bottom resistor → output

The combined resistance is the sum of the individual resistances.

5. **Both resistors in parallel**.

i.e. input → top resistor, input → bottom resistor,

top resistor → output, bottom resistor → output

The combined resistance is less than either of the individual resistances.

For the curious... it's: $1 / (1 / (\text{top resistance}) + 1 / (\text{bottom resistance}))$

Don't worry, this won't be on the test!

Reading resistors

Each resistor has a sequence of three colored bands, indicating a two-digit number and a multiplier. A fourth band indicates a tolerance value (not used).

The fourth band is separated by a gap from the first three. Resistors can be rotated; take care to read the bands in the correct direction.

Color	First Band	Second Band	Multiplier
Black	0	0	1Ω
Brown	1	1	10Ω
Red	2	2	100Ω
Orange	3	3	$1,000\Omega$
Yellow	4	4	$10,000\Omega$
Green	5	5	$100,000\Omega$
Blue	6	6	$1,000,000\Omega$
Violet	7	7	$10,000,000\Omega$
Gray	8	8	—
White	9	9	—
Gold	—	—	0.1Ω
Silver	—	—	0.01Ω

For example, **Green Violet Yellow** indicates $57 \times 10,000\Omega = 570,000\Omega$.

On the Subject of Cryptography

WLMY ETGXFD EQCD ED PQKW WT CMFF EZYDFB.

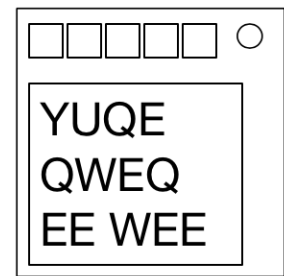
SEE APPENDIX CD43 FOR AN EXCERPT OF "A CHRISTMAS CAROL".

SEE APPENDIX CD44 FOR FREQUENT LETTERS AND WORDS.

This module will display ciphertext which contains a sentence from Charles Dickens' "A Christmas Carol" (aka the plaintext). The plaintext has been encrypted via a substitution cypher, meaning each letter in the alphabet is substituted for a different letter.

- The letter E will always mean the letter E.
- Apart from the letter E, no letter can substitute itself.
- All punctuation has been removed from the ciphertext.
- Above the display are five keys each with a letter that is found in the plaintext.
- Once the ciphertext is decrypted, press each key only once, in order that they appear in the plaintext.
- Entering the incorrect sequence will gain a strike, try the sequence again from the beginning.

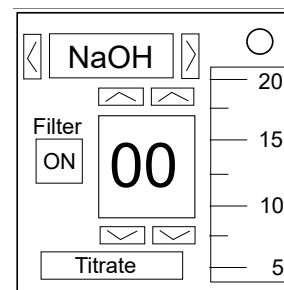
NOTE: The meaning of the word colors is currently unknown, however you can safely ignore them.



On the Subject of Neutralization

The rules are simple: neutralize or be neutralized.

- The module is disarmed by successfully neutralizing an acid contained in a tube by titrating it with a chemical base.
- In order to solve the module, the type of base, amount of base, and filter state must all be correct.
- Once the appropriate conditions are set, press “Titrate” to confirm the solution.
- An incorrect input yields a strike. The correct answer remains unchanged.
- Useful info may be found in **Appendix NT27: Chemical Information**.



Determining Titants

The acid type can be determined using the following chart:

<u>Solution Color</u>	<u>Acid Type</u>
Red	Hydrogen bromide
Yellow	Hydrogen fluoride
Green	Hydrogen chloride
Blue	Hydrogen iodide

The base that must be used to titrate can be determined via the following ruleset:

- If the bomb has an NSA indicator and exactly 3 batteries, add ammonia.
- Otherwise, if the bomb has a lit CAR, FRQ, or IND indicator, add potassium hydroxide.
- Otherwise, if the bomb has no ports and the serial number has a vowel, add lithium hydroxide.
- Otherwise, if the acid's chemical formula has a letter in common with an indicator present on the bomb, add potassium hydroxide.
- Otherwise, if the number of D batteries is greater than the number of AA batteries, add ammonia.
- Otherwise, if the anion's atomic number is less than 20, add sodium hydroxide.
- Otherwise, add lithium hydroxide.

Determining Concentrations

The concentration of the acid can be determined via the following process:

- Start with the atomic number of the anion of the acid.
- Subtract the atomic number of the cation of the base.
- If the anion or cation has a vowel in the chemical symbol, subtract 4.
- If the anion and cation's chemical symbols have the same number of characters, multiply by 3.
- Take the least significant digit of the result (removing negative signs).
- If the number is 0, the number becomes the volume of acid doubled then divided by 5.
- Divide by 10. This is the concentration of the acid.

The concentration of the base can be determined via the following ruleset:

- If there are more battery holders than port types and more battery holders than indicators, the concentration is 5.
- If there are more port types than battery holders and more port types than indicators, the concentration is 10.
- If there are more indicators than battery holders and more indicators than port types, the concentration is 20.
- If there are any ties for the most, the concentration is either 5, 10, and 20, whichever is closest to the cation's atomic number.
- However, if the titration combination is HI and KOH or HCl and NH₃, the concentration is always 20.

Determining Drop Count

- Start with 20 and divide by the concentration of the base.
- Multiply by the volume of acid and concentration of the acid.
- The result is the number of drops required to successfully titrate.

Determining Solubility

- If the module's acid/base combination on the following chart has "NS" for "Not Soluble", the filter must be turned ON before the base is added.
- Otherwise, the filter must be turned OFF.

	<u>NH₃</u>	<u>KOH</u>	<u>LiOH</u>	<u>NaOH</u>
<u>HBr</u>	S	NS	NS	S
<u>HF</u>	NS	S	NS	S
<u>HCl</u>	NS	NS	S	NS
<u>HI</u>	S	S	S	NS

APPENDIX NT27: Chemical Information

NT27.1: Bases

<u>Name</u>	<u>Chemical Formula</u>	<u>Cation</u>	<u>Chemical Symbol</u>	<u>Atomic Number</u>
Ammonia	NH ₃	Hydrogen	H	1
Lithium hydroxide	LiOH	Lithium	Li	3
Sodium hydroxide	NaOH	Sodium	Na	11
Potassium hydroxide	KOH	Potassium	K	19

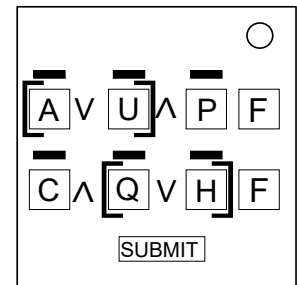
NT27.2: Acids

<u>Name</u>	<u>Chemical Formula</u>	<u>Anion</u>	<u>Chemical Symbol</u>	<u>Atomic Number</u>
Hydrofluoric acid	HF	Fluorine	F	9
Hydrochloric acid	HCl	Chlorine	Cl	17
Hydrobromic acid	HBr	Bromine	Br	35
Hydroiodic acid	HI	Iodine	I	53

On the Subject of Logic

Logic is easy, but logic AND bomb defusal might not.

- Each row displays 3 letters. Each letter represents a statement which can be found in table 1.
- On each row, solve the statements inside the brackets first.
- Statements are joined with logical connective symbols. Find how each symbol works in table 2.
- Apply negation (NOT gate: false becomes true and true becomes false) to each statement first if the red LED above that statement is lit.
- Find the end result of each row, and then use the T/F button to the right to select True/False. Press “Submit” when done.



See Appendix A for indicator identification reference.

See Appendix B for battery identification reference.

See Appendix C for port identification reference.

Table 1: Statement list

Letter	Statement	Letter	Statement
A	Number of batteries = number of indicators	N	More than 2 battery holders
B	Serial number has more letters than digits	O	Has both lit and unlit indicators
C	Has IND indicator	P	Has parallel port
D	Has FRK indicator	Q	Exactly 2 ports
E	Exactly 1 unlit indicator	R	Has PS/2 port
F	More than 1 port type	S	Sum of digits in serial number > 10
G	2 batteries or more	T	Has MSA indicator
H	Less than 2 batteries	U	Exactly 1 battery holder
I	Last digit of serial number is odd	V	Serial number contains vowels
J	More than 4 batteries	W	No indicators
K	Exactly 1 lit indicator	X	Exactly 1 indicator
L	More than 2 indicators	Y	More than 5 ports
M	No duplicate ports	Z	Less than 2 ports

Table 2: Logical connective symbol list

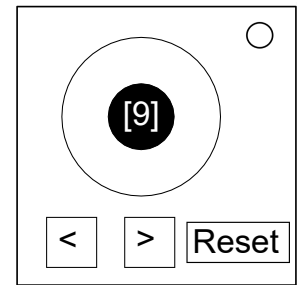
Logical Connective	Symbol	Logic Gate Equivalent	Meaning
Conjunction	$\wedge$	AND	Returns true if all inputs are true. Else returns false.
Disjunction	$\vee$	OR	Returns true if any input is true. Else returns false.
Exclusive Disjunction	$\veebar$	XOR	Returns true if exactly one input is true. Else returns false.
Alternative Denial	$\downarrow$	NAND	Returns false if all inputs are true. Else returns true.
Joint Denial	$\uparrow$	NOR	Returns false if any input is true. Else returns true.
Biconditional	$\leftrightarrow$	XNOR	Returns false if exactly one input is true. Else returns true.
Implication (Left)	$\rightarrow$	-	Returns false when left input is true and right input is false. Else returns true.
Implication (Right)	$\leftarrow$	-	Returns false when left input is false and right input is true. Else returns true.

On the Subject of Combination Locks

This looks like a combination lock. I thought I was disarming this bomb, not unlocking it.

See Appendix B of original manual for battery identification reference.

See Appendix Two Factor for two factor identification reference.



Like a typical combination lock, this requires 3 numbers to unlock. Turn the dial to the right to the first number. Then turn it to the left for the second number. Finally, turn it to the right for the last number. That will unlock it!

If sequential numbers in the code are the same, perform a full revolution back to the same number.

How to decode the combination:

Each number ranges from 0-19. Refer to the table to determine each number. Be careful if there are two factor codes present because they change periodically!

First number:

1. Add the least significant digit of each two factor code together.
2. If there are no two factor codes, use the last digit of the serial number plus the number of solved modules.
3. Add the number of batteries.
4. Subtract 20 if the result is over 19.

Second number:

1. Add the most significant digit of each two factor code together.
2. If there are no two factor codes, use the number of modules on the bomb (including needy modules).
3. Add the number of solved modules.
4. Subtract 20 if the result is over 19.

Third number:

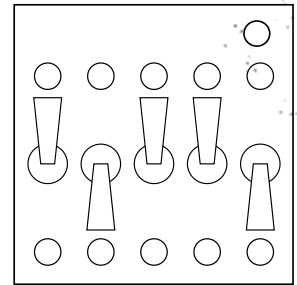
1. Add the first two numbers together.
2. Subtract 20 if the result is over 19.

On the Subject of Switches

A yes or no choice isn't too bad. Unfortunately you have to make five of them and any of them could be your last.

Switches need to be flipped to match the lit indicators either above or below them.

Avoid the following switch states:



Section 2: Needy Modules

Needy modules cannot be disarmed, but pose a recurrent hazard.

Needy modules can be identified as a module with a small 2-digit timer in the top center. Interacting with the bomb may cause them to become activated. Once activated, these needy modules must be tended to regularly before their timer expires in order to prevent a strike.

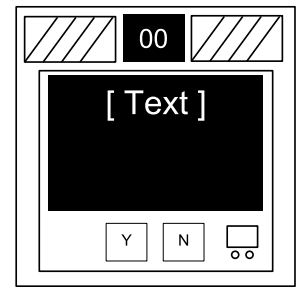
Stay observant: needy modules may reactivate at any time.

00

On the Subject of Venting Gas

Computer hacking is hard work! Well, it usually is. This job could probably be performed by a simple drinking bird pressing the same key over and over again.

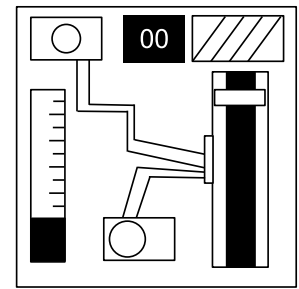
- Respond to the computer prompts by pressing "Y" for "Yes" or "N" for "No".



On the Subject of Capacitor Discharge

I'm going to guess that this is just meant to occupy your attention, because otherwise this is some shoddy electronics work.

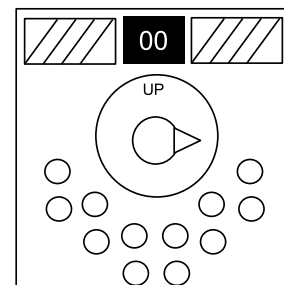
- Discharge the capacitor before it overloads by holding down the lever.



On the Subject of Knobs

Needlessly complicated and endlessly needy. Imagine if such expertise were used to make something other than diabolical puzzles.

- The knob can be turned to one of four different positions.
- The knob **must** be in the correct position when this module's timer hits zero.
- The correct position can be determined by the on/off configuration of the twelve LEDs.
- Knob positions are relative to the "UP" label, which may be rotated.



LED Configurations

Up Position:

		X		X	X
X	X	X	X		X

X		X		X	
	X	X		X	X

Down Position:

	X	X			X
X	X	X	X		X

X		X		X	
	X				X

Left Position:

				X	
X			X	X	X

				X	
			X	X	

Right Position:

X		X	X	X	X
X	X	X		X	

X		X	X		
X	X	X		X	

X = Lit LED

On the Subject of Hex To Decimal

Be happy you are only doing this for one game session. At least you weren't Nintendo in the 1980's figuring out how to program the NES with 0's and 1's.

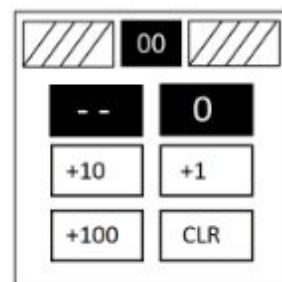
The main display at the top left displays a random hexadecimal character.

The input display at the top right displays the number the defuser has currently added.

Starting from 0, the defuser must add 100, 10, or 1 to get the desired number.

If the defuser messes up, reset by pressing the "CLR" button.

• To find the desired number, use the table below:



	-0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-A	-B	-C	-D	-E	-F
0-	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1-	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
2-	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
3-	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
4-	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79
5-	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95
6-	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111
7-	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127
8-	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143
9-	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159
A-	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175
B-	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191
C-	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207
D-	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223
E-	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239
F-	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255

On the Subject of HTTP Response

When your bomb acts as a web server...

- The module displays a remote or local status string. You must response to it with a 3-digit status code.
- Remote codes can be found on Table 1 while local codes can be found on Table 2.
- An undisclosed source told us that, due to display limits, some strings may be abbreviated.
- Use the keypad to enter the correct response code to disable the needy module.
- **Caution:** There's no delete button, so be careful!

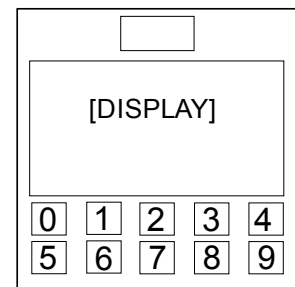


Table 1: Remote status, response as listed.

Full name	Abbrev.	Response	Full name	Abbrev.	Response
Continue	Cont	100	Unauthorized	Unauth	401
Switching Protocols	SwPrt	101	Payment Required	PayReq	402
Processing	Proc	102	Forbidden	Frbd	403
OK	OK	200	Not Found	NFnd	404
Created	Crtd	201	Request Time-out	TmOut	408
Moved Permanently	MvPerm	301	Gone	Gone	410
Found	Found	302	I'm a teapot	ImaTp	418
Not Modified	NMod	304	Internal Server Error	SrvErr	500
Use Proxy	UPrx	305	Bad Gateway	BGw	502
Bad Request	BadR	400	Service Unavailable	SrvUn	503

Table 2: Local status. Add the number of batteries times the sum of the numerical digits in the serial number to the response first.

Full name	Abbrev.	Response	Fullname	Abbrev.	Response
Strike	Stk	601	Detonate	Dtn	602
Pass	Ps	603	Press	Prs	604
Command	Cmd	605	Release	Rls	606
Blue	Blue	701	Green	Grn	702
Red	Red	703	Yellow	Yel	704
Black	Blk	705	White	Wht	706
Cut	Cut	801	Unscrew	Uns	802
Set	Set	803	Locate	Loc	804
Indicator	Indc	805	Batteries	Batt	806
Timer	Tmr	901	Lower	Lwr	902
Upper	Upr	903	Lever	Lvr	904
Time	Time	905	Done	Done	906

Note: In the rare case the response number is over 999, subtract 999 from the response code.

Appendix A: Indicator Identification Reference

Labelled indicator lights can be found on the sides of the bomb casing.

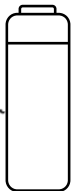
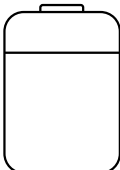


Common Indicators

- SND
- CLR
- CAR
- IND
- FRQ
- SIG
- NSA
- MSA
- TRN
- BOB
- FRK

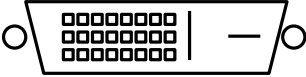
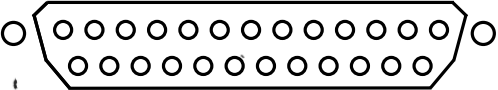
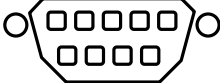
Appendix B: Battery Identification Reference

Common battery types can be found within enclosures on the sides of the bomb casing.

Battery	Type
	AA
	D

Appendix C: Port Identification Reference

Digital and analog ports can be found on sides of the bomb casing.

Port	Name
	DVI-D
	Parallel
	PS/2
	RJ-45
	Serial
	Stereo RCA

Appendix: Grammar

for use with the English Test module

This appendix contains a brief overview of some grammatical distinctions used in the English Test module and Needy English Test module.

The **subject** is what is doing the action, and the **object** is what is receiving the action. e.g. In "I buy milk." I is the subject and milk is the object.

their: belonging to them; there: that place; they're: they are
your: belonging to you; you're: you are
I, he, she, we, they: used in subjects; me, him, her, us, them: used in objects
less: used with uncountable nouns; fewer: used with countable nouns
who: used in subjects; whom: used in objects
defiantly: rebelliously; definitely: without doubt
lead: the metal or the present tense; led: the past tense and past participle
cite: declare a quoted source; site: location; sight: a view or vision
When you don't lay something else down, you lie down. The past tense of <u>lay</u> is laid . Confusingly, the past tense of <u>lie</u> is lay !
Literally means <u>word for word</u> . If you had "literally died" watching a video, your family and friends would be crying at your funeral about now.
If you write "should of ", "could of ", "would of ", or "might of ", no educated gentleman will take you seriously. Remember, "I <u>do</u> " is to "I <u>have done</u> " as "I could <u>do</u> " is to "I could <u>have done</u> ". (Exceptions apply, but very <u>very</u> rarely!)
its: belonging to it; it's: it is
capital: main city in a territory, money you put up to borrow something, or THIS KIND OF LETTER; capitol: big building, usually in a <u>capital</u>
affect: usually a verb, but noun when it means "display of emotion"; effect: almost always a noun; impact: physical force
i.e.: short for Latin <i>id est</i> , or "that is"; e.g.: short for Latin <i>exempli gratia</i> , or "for example"
peak: summit; peek: sneak a look; poke: excite (usually interest)
allot: partition; a lot: very much; alot: (never correct)
lose: opposite of gain; loose: opposite of tight
than: (used to compare two things); then: at the time, or right after that
complement: when two parts complete each other; compliment: You look good today!
farther: physical distance; further: figurative distance
number: used for countable nouns; amount: used for uncountable nouns

Appendix: Grammar (Cont'd)

to: used in infinitives or destination; too: as well, or overly; two: 2
accept: This is fine; except: One of these things is not like the others
threw: past tense of "throw"; through: in at one side/end and out at the other
defuse: stop a bomb; diffuse: light softening out. Use "defuse" for tension.
statue: monument; stature: body height; statute: code of law
stationary: completely still; stationery: writing utensils
by: beside, from the mind of, etc.; buy: trade money for goods; bye: see you later
breath: the noun; breathe: the verb
drink: present tense; drank: past tense; drunk: past participle and adjective
discreet: secret or carefully subtle; discrete: separate
seas: plural of sea; sees: a form of "to see"; seize: to grab or take by force; C's: more than one C
weather: condition of the outside air; whether: if it is or if it isn't
raise: to make something go up; rays: narrow beams of light; raze: get rid of hair with a razor, or similarly destroy a wide area
wander: frolic; wonder: ponder
die: stop living, or a small cube for randomness; dice: more than one die
meat: flesh; meet: to see someone else; mete: to deal out something unpleasant
palate: roof of your mouth; palette: board to mix paint on, or a combination of colors; pallet: plates that cargo gets placed on
In this module, racket: a loud noise; racquet: a netted stick or paddle with which to hit a ball. (Especially in US English, racket can be used for both senses.)
perfect: 100% good or correct; prefect: person in a position of power, like an official or a heir

Other pairs/sets of words include: ad/add, aloud/allowed, altar/alter, arc/ark, baited/bated, base/bass, blew/blue, brake/break, carat/caret/carrot/karat, ceiling/sealing, cent/scent/sent, cereal/serial, choral/coral/corral, coarse/course, creak/creek, dear/deer, discussed/disgust, elicit/illicit, everyday/every day, faint/feint, faze/phase, find/fined, flair/flare, flea/flee, gait/gate, idle/idol/idyll, lighting/lightning/lightening, loan/lone/lend, oar/or/ore, pail/pale, pair/pare/pear, poor/pore/pour, praise/prays/preys, precedence/precedents/presidents, right/rite/wright/write, road/rode/rowed, ring/wring, role/roll, seam/seem, stairs/stares, steal/steel, straight/strait, though/thought/through/thorough, vain/vane/vein, vary/very, wait/weight, and weak/week.

For lack of space, the differences for these words have been omitted, but they should be fairly common knowledge to most English speakers.

APPENDIX CD43

Excerpt from Charles Dickens' "A Christmas Carol".

Scrooge knew he was dead? Of course he did. How could it be otherwise? Scrooge and he were partners for I don't know how many years. Scrooge was his sole executor, his sole administrator, his sole assign, his sole residuary legatee, his sole friend, and sole mourner. And even Scrooge was not so dreadfully cut up by the sad event, but that he was an excellent man of business on the very day of the funeral, and solemnised it with an undoubted bargain. The mention of Marley's funeral brings me back to the point I started from. There is no doubt that Marley was dead. This must be distinctly understood, or nothing wonderful can come of the story I am going to relate. If we were not perfectly convinced that Hamlet's Father died before the play began, there would be nothing more remarkable in his taking a stroll at night, in an easterly wind, upon his own ramparts, than there would be in any other middle-aged gentleman rashly turning out after dark in a breezy spot -- say Saint Paul's Churchyard for instance -- literally to astonish his son's weak mind.

Scrooge never painted out Old Marley's name. There it stood, years afterwards, above the warehouse door: Scrooge and Marley. The firm was known as Scrooge and Marley. Sometimes people new to the business called Scrooge Scrooge, and sometimes Marley, but he answered to both names. It was all the same to him.

Oh! But he was a tight-fisted hand at the grind-stone, Scrooge! A squeezing, wrenching, grasping, scraping, clutching, covetous, old sinner! Hard and sharp as flint, from which no steel had ever struck out generous fire; secret, and self-contained, and solitary as an oyster. The cold within him froze his old features, nipped his pointed nose, shrivelled his cheek, stiffened his gait; made his eyes red, his thin lips blue and spoke out shrewdly in his grating voice. A frosty rime was on his head, and on his eyebrows, and his wiry chin. He carried his own low temperature always about with him; he iced his office in the dogdays; and didn't thaw it one degree at Christmas.

External heat and cold had little influence on Scrooge. No warmth could warm, no wintry weather chill him. No wind that blew was bitterer than he, no falling snow was more intent upon its purpose, no pelting rain less open to entreaty. Foul weather didn't know where to have him. The heaviest rain, and snow, and hail, and sleet, could boast of the advantage over him in only one respect. They often 'came down' handsomely, and Scrooge never did.

Nobody ever stopped him in the street to say, with gladsome looks, 'My dear Scrooge, how are you? When will you come to see me?' No beggars implored him to bestow a trifle, no children asked him what it was o'clock, no man or woman ever once in all his life inquired the way to such and such a place, of Scrooge. Even the blind men's dogs appeared to know him; and when they saw him coming on, would tug their owners into doorways and up courts; and then would wag their tails as though they said, 'No eye at all is better than an evil eye, dark master!'

But what did Scrooge care! It was the very thing he liked. To edge his way along the crowded paths of life, warning all human sympathy to keep its distance, was what the knowing ones call 'nuts' to Scrooge.

APPENDIX CD44Word and Letter Frequency in Charles Dickens' "A Christmas Carol".**Frequent 2 letter words**

- 16x TO
- 11x NO
- 10x HE
- 9x IN
- 8x OF
- 7x IT

Frequent 3 letter words

1. 23x AND
2. 22x THE
3. 22x HIS
4. 14x WAS
5. 11x HIM
6. 4x OUT

Frequent 4 letter words

1. 6x SOLE
2. 4x THAT

Frequent 5 letter words

1. 4x THERE
2. 4x WOULD

Frequent 6 letter words

1. 4x MARLEY

Frequent 7 letter words

1. 12x SCROOGE

Frequent 8 letter words

1. 2x BUSINESS

Frequent 9 letter words

1. 2x SOMETIMES